



NSAT Phase-1 Question Paper (1)

Sr. No.	Section Name		No. of Questions
1	General Aptitude	Logic & Data Interpretation	20
		Algorithmic Thinking	10
2	English	Reading Comprehension	10
		Language Reasoning	10
3	Mathematics	10 th Class Maths	20
		11 th & 12 th Class Maths	10



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SECTION – 1: GENERAL APTITUDE

Directions for questions 1 – 3: Read the given information and answer the following questions.

In a small forest, there are five animals: Monkey, Hare, Elephant, Bear and Fox, whose names are Nikky, Chikky, Rikky, Mikky and Dikky, not in the same order. Each of them likes to eat a different fruit available in the forest among Berry, Cherry, Ferry, Merry and Terry, not in the same order.

It is known that:

1. The Fox's favourite fruit is Terry and its name is not Rikky.
2. Dikky likes Merry.
3. Nikky is a Hare and Monkey's name is Chikky.
4. Monkey does not like Ferry and the Elephant's favourite fruit is Cherry.

1. Who likes Ferry?

- (1) Monkey (2) Hare (3) Elephant (4) Bear
(5) None of these

2. Whose name is Rikky?

- (1) Monkey (2) Hare (3) Elephant (4) Bear
(5) None of these

3. What is the favourite fruit of the Monkey?

- (1) Berry (2) Cherry (3) Merry (4) Terry
(5) None of these

4. Ritika went for shopping one evening. She found that Jaisika was coming from the opposite direction and the shadow of Jaisika was exactly to the left of Ritika. Which direction would Jaisika face, if she took a turn of 90° clockwise after meeting Ritika?

- (1) East (2) South-west (3) North-west (4) West (5) North-east

5. Ranjit started his journey towards south; and after travelling 10 km, he took a right turn and moved 5 km. Then he took a right turn and moved 2 km. He again took a right turn and travelled 15 km. Finally, he took a left turn and travelled 32 km and stopped. How far was he from the starting position?

- (1) 20 km (2) 22 km (3) 23 km (4) 26 km
(5) None of above

Directions: Read the following information carefully and answer the question given below.

'A + B' means 'A is the sister of B'.

'A - B' means 'A is the father of B'.

'A / B' means 'A is the brother of B'.

'A * B' means 'A is the daughter of B'.

'A % B' means 'A is the wife of B'.



6. If we have A * E * I - M - Q, then which of the following is correct?
- (1) I is the paternal grandfather of A. (2) I is the maternal grandfather of Q.
(3) E is the maternal aunt of Q. (4) M is the paternal uncle of A.
(5) None of these
7. Eight people - P, Q, R, S, T, U, V and K - are sitting on a bench for a photograph. V is the son of R and has 2 sisters. S, who is the son of T, is the brother-in-law of V. Q, who is the daughter-in-law of P, has a sister U. T is the grandmother of Q's son K. V is unmarried.
- How is V related to K?
- (1) Father (2) Cousin (3) Uncle (4) Brother
(5) None of above
8. In a coding system, if FURNITURE is coded as [23372332733225372335] and GLOSS is coded as [7223351919], then BOTTLE will be coded as
- (1) [2352252252335] (2) [2252252252235] (3) [2252252252335]
(4) [2352252252235] (5) None of above

Directions for questions 9 – 10: Study the given information and answer the following questions.

In a certain code language,

"Never dreamed about success" is written as "4D 1T 19S 14R".

"Challenges are forced gifts" is written as "1E 7S 6D 3S".

"Sometimes smallest decisions change" is written as "19S 4S 3E 19T".

9. What will be the code of 'believe' in the given code language?
- (1) 2B (2) 2E (3) 5E (4) 5C
(5) None of these
10. If 'Hope for _____' is coded as '8E 6R 20W' in the given code language, then what will be the missing word?
- (1) cashew (2) tomorrow (3) hacksaw (4) preview
(5) None of above

Directions for questions 11 – 13:

Study the given information and answer the following questions. The table below shows the distribution of students in an institution in three grades - Average, Good and Outstanding - across three levels - Primary, Secondary and Higher Secondary - on the basis of their performance in the final examination over a period of six years beginning 2012.



Level	Primary			Secondary			Higher Secondary		
Grade	Average	Good	Outstanding	Average	Good	Outstanding	Average	Good	Outstanding
2012	120	75	35	100	68	24	98	82	29
2013	115	68	34	98	75	22	89	78	37
2014	132	54	30	105	60	25	95	90	29
2015	145	64	25	110	70	28	102	75	34
2016	128	60	28	125	78	18	99	88	28
2017	110	80	32	120	56	21	110	90	32

11. The ratio of the number of average students in the year 2012 to the number of above average students in the year 2016 is

- (1) 21 : 20 (2) 53 : 50 (3) 40 : 21 (4) 31 : 30
(5) None of above

12. For which year was the percentage of students securing a good grade the highest among students of secondary level?

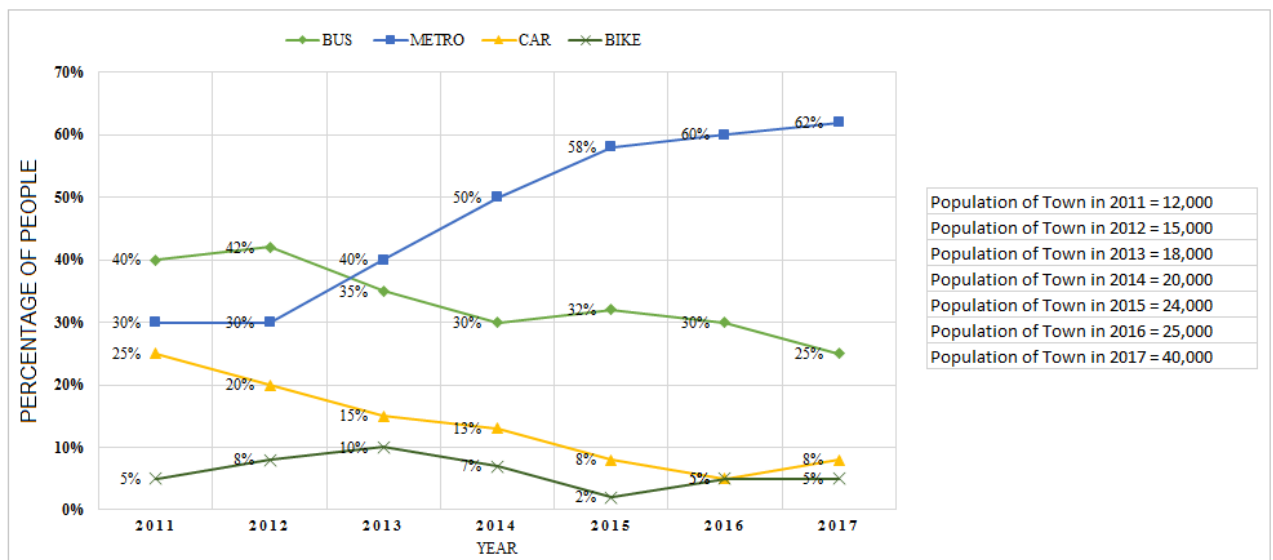
- (1) 2013 (2) 2015 (3) 2017 (4) 2016
(5) None of above

13. Which year had the highest student strength?

- (1) 2015 (2) 2013 (3) 2016 (4) 2014
(5) None of above

Directions for questions 14 – 16: Study the following graph carefully and answer the questions.

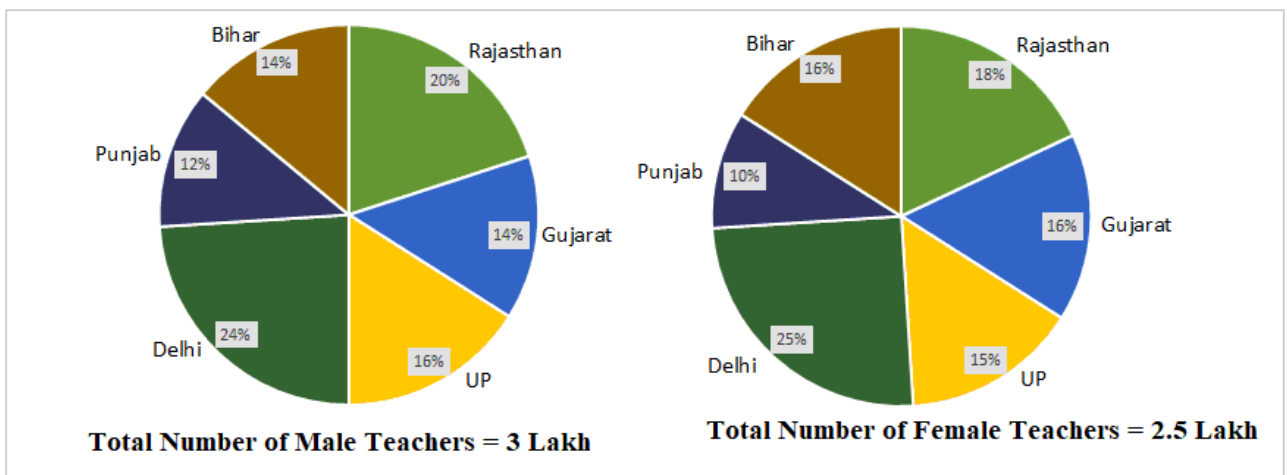
Preference of People (in Percentage) In Using Different Modes of Transport Over the Years





14. What is the ratio of the number of people who preferred to travel by bike in 2011 to the number of people who preferred to travel by bus in 2017?
(1) 50 : 3 (2) 3 : 50 (3) 30 : 7 (4) 5 : 7
(5) None of these
15. What is the difference between the total number of people who preferred to travel by bus in 2012 and 2015 together and the total number of people who preferred to travel by metro in 2014 and 2017 together?
(1) 20,220 (2) 22,200 (3) 20,820 (4) 20,200
(5) None of these
16. By what percentage is the number of people who preferred to travel by metro in the year 2017 greater than the number of people who preferred to travel by bus in 2015?
(1) 23% (2) 123% (3) 220% (4) 223%
(5) None of these

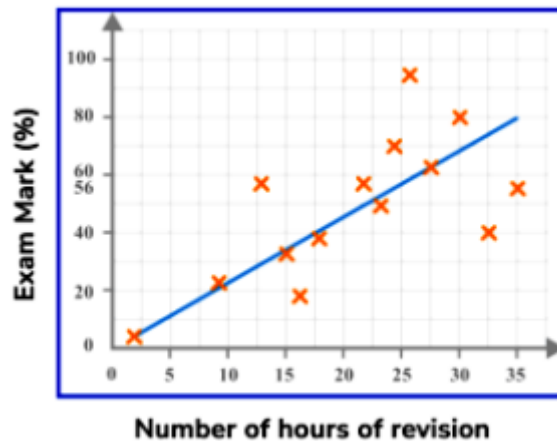
Directions for questions 17 – 19: Study the following pie-chart carefully and answer the questions.



17. The total number of male teachers from Rajasthan, Bihar and Gujarat together is what percentage of the total number of female teachers from UP, Delhi and Punjab together?
(1) 115.2% (2) 125.5% (3) 105.4% (4) 140.6%
(5) None of these
18. If the sum of total number of teachers (male + female) from Delhi and total number of teachers (male + female) from UP is A, and the sum of total number of teachers (male + female) from Bihar and the total number of teachers (male + female) from Punjab is B, then the value of (A + B) is
(1) 1,43,000 (2) 2,20,000 (3) 3,63,000 (4) 5,63,000
(5) None of these
19. The total number of male teachers from Delhi, UP and Rajasthan together is how much more or less than the total number of female teachers from Bihar, Gujarat and Punjab together?
(1) 85,000 less (2) 75,000 more (3) 65,000 more (4) 55,000 less
(5) None of these



Directions: This scatter plot examines the correlation between the hours of revision for exam preparation and the subsequent exam marks represented as percentage.



20. If the maximum marks of the exam are 400, what are the possible marks obtained by the student who has revised for 15 hours?
- (1) 40 (2) 100 (3) 140 (4) 200 (5) 240

Directions for questions 21 – 24:

Study the following information carefully and answer the questions that follow.

The Indian Air Force (IAF) Recruitment Board is sending out call letters to eligible male and female Indian citizens for the final interview for admission to the courses commencing in 2017 for grant of Short Service Commission in Flying Branch. Candidates must satisfy the following criteria to be eligible for an interview call.

1. **Age:** The candidate must be 20-24 years old (as on 1st January, 2016).
2. **Educational Qualifications:** Candidates must satisfy criteria 2.1 and 2.2 or 2.3.
 - 2.1: Physics, Chemistry and Mathematics at 10 + 2 level with a minimum of 70% marks in Physics, Mathematics and Chemistry each
 - 2.2: Graduates from Science (Physics and Mathematics majors) discipline from a recognised university, who have attained a minimum of 65% marks in aggregate in all papers
 - 2.3: B.E. or B.Tech. (Mechanical, Aerospace, Mechanical Automation or Mechatronics) with minimum 65% marks
3. **Physical Standards:**
 - 3.1: Minimum height 165 cm for both men and women with correlated weight, subject to a minimum weight of 60 kg for men and 58 kg for women
 - 3.2: Maximum height 190 cm for both men and women with correlated weight, subject to a maximum weight of 95 kg for men and 85 kg for women
 - 3.3: Vision - Minimum Visual Acuity 6/6 in one eye and 6/9 in the other, correctable to 6/6 only
4. **Marital Status:** Unmarried
5. **AFCAT Score:** The minimum score requirement is 60% for an interview call.

Following exemptions are valid for candidates with below mentioned attributes.



Exemption to criterion 1: Candidates holding a valid and current Commercial Pilot Licence issued by DGCA (India) may have their age relaxed by 5 years, subject to a maximum of 29 years, in which case their suitability for eligibility shall be reviewed by the approval committee and duly notified accordingly.

Exemption to criterion 2.1: Candidates who do not meet criterion 2.1 but meet criterion 5 and were sportspersons at school level get an exemption of 5% as set in criterion 2.1, in which case their case shall be forwarded to the selection committee.

Exemption to criteria 2.2 and 2.3: Candidates who do not meet criterion 2.2 or 2.3 but meet criterion 5 and were sportspersons at university level get an exemption of 5% as set in criterion 2.2 or 2.3, in which case their case shall be forwarded to the selection committee.

Exemption to criterion 5: Candidates who have scored 75% or more as set in criterion 2.2 or 2.3 get an exemption of 5% in their AFCAT score as set in criterion 5, in which case their case shall be forwarded to the selection committee.

21. Abhay Singh Chauhan's date of birth is 30th June, 1993. He had scored 100% marks in Physics and Mathematics each and 80% marks in Chemistry at 10 + 2 level. He passed his graduation with B.Tech. in Aerospace from PEC, Chandigarh with 80% marks. He was a body-builder in his university days and was placed 3rd in the North zone inter-university competition in 2014. He is 180 cm tall with a body weight of 85 kg and with perfect 6/6 vision. He is currently single. He has scored 75% marks in the AFCAT.

Which of the following actions should be performed by the IAF Recruitment Board?

- (1) Forward the case to the approval committee for review and notification.
- (2) Call the candidate for an interview.
- (3) Do not call the candidate for an interview.
- (4) Forward the case to the selection committee.
- (5) Data insufficient

22. Gurmeher Kaur is 28 years old as on 1st January, 2016. She is currently working as a pilot in Jet Airways, but her adventurous spirit yearns for a more action packed lifestyle and hence, Air Force. She had scored 95% marks in class 10 + 2 with 100 each in Physics, Chemistry and Mathematics. She is a graduate with B.Tech. in Computer Science (Thapar University) and was a university topper with 85% marks. She has been a national level swimmer and is 175 cm tall weighing 75 kg. She is endowed with perfect vision. She is currently unmarried.

Which of the following will be the outcome of her application?

- (1) Data insufficient
- (2) Her case gets forwarded to the selection committee.
- (3) She is called for an interview.
- (4) Her case is under review and pending for notification by the approval committee.
- (5) She is not called for an interview.

23. Rajkumar Verma was 22 years old as on 1st January, 2016. He is currently single. He had scored 80% marks in 10 + 2 with 95% in Physics, 85% in Mathematics and 75% in Chemistry. He scored 64% marks in his B.Sc. with Physics and Mathematics as majors. He was the soccer captain of his university soccer team, which



won the zonal university soccer championship. He is 180 cm tall with a body weight of 82 kg. His vision is perfectly normal as 6/6 in both eyes. He has scored 72% in the AFCAT exam.

Which of the following would be the outcome of his case?

- (1) He is called for an interview.
- (2) He is not called for an interview.
- (3) His case is under review and pending for notification by the approval committee.
- (4) His case is forwarded to the selection committee.
- (5) Data insufficient

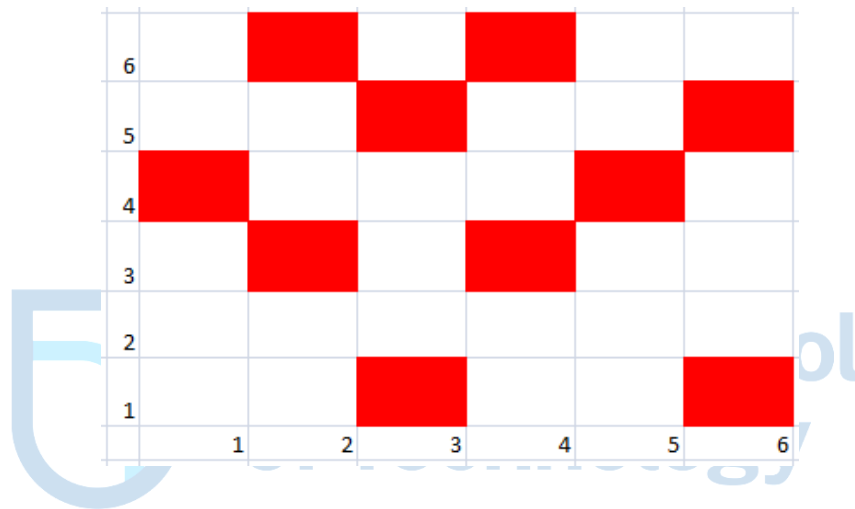
24. Swati Singh celebrated her 23rd birthday on 31st December, 2015 and took a vow to join the Air Force just like her dad. She passed her B.Sc. honours from St. Stephens New Delhi with Physics and Mathematics as majors with an aggregate of 85%. She topped the Delhi zone in her class XII exams having scored 98.5% aggregate with 100% in Physics and Mathematics each. She is 170 cm tall with a body weight of 70 kg. She was a national level shooting champion in her university and is trying to qualify for the Indian Olympics team. She is currently single. She scored 80% in the AFCAT exam.

What should be the outcome of her case?

- (1) Ineligible
- (2) Cleared for the interview round
- (3) Case forwarded to the approval committee
- (4) Case forwarded to the selection committee
- (5) Data insufficient

Directions for questions 25 – 28: Read the following information carefully to answer the given questions.

A drone is deployed in a complex airspace configuration represented by a grid. The drone is equipped with the ability to execute instructions: MOVE FORWARD, TAKE LEFT TURN, TAKE RIGHT TURN. The airspace grid is comprised of cells, each denoting either a clear flight path or an airspace restriction. The drone embarks on its mission from a specific starting point and is initially oriented in one of the cardinal directions (North, East, South, West). The mission objective is to guide the drone through the airspace grid to reach a predetermined destination. The drone initiates its journey from cell coordinates (2, 3) in the grid, facing North. The airspace layout is illustrated as a 6 x 6 grid, where each cell is assigned a unique characteristic. Blank cells signify open flight paths, while red cells indicate restricted airspace. The drone is capable of moving one step at a time from its current position. Cell coordinates (a, b) define a position in terms of the bth column from the left and ath row from the bottom.



25. After starting at (2, 3) and moving forward, moving forward again, and making another left turn, what are the final coordinates of the drone?
- (1) (2, 4) (2) (4, 2) (3) (3, 1) (4) (5, 3)
- (5) None of these
26. If the drone is at (1, 4) and facing North and wants to reach (5, 4), what is the minimum number of instructions that it executes?
- (1) 4 (2) 6 (3) 8 (4) 10
- (5) None of these
27. To reach cell coordinates (3, 1) from point (4, 2) facing East, which set of instructions should the drone follow?
- (1) TAKE RIGHT TURN, MOVE FORWARD, MOVE FORWARD, TAKE RIGHT TURN, MOVE FORWARD, MOVE FORWARD, TAKE RIGHT TURN, MOVE FORWARD, MOVE FORWARD
- (2) MOVE FORWARD, TAKE LEFT TURN, MOVE FORWARD, MOVE FORWARD, TAKE RIGHT TURN, MOVE FORWARD, MOVE FORWARD, TAKE LEFT TURN, MOVE FORWARD
- (3) MOVE FORWARD, TAKE LEFT TURN, MOVE FORWARD, MOVE FORWARD, TAKE LEFT TURN, MOVE FORWARD, MOVE FORWARD, TAKE RIGHT TURN, MOVE FORWARD
- (4) MOVE FORWARD, TAKE RIGHT TURN, MOVE FORWARD, MOVE FORWARD, TAKE RIGHT TURN, MOVE FORWARD, MOVE FORWARD, TAKE RIGHT TURN, MOVE FORWARD
- (5) None of these
28. If the drone develops a fault and happens to automatically execute the instruction "TAKE LEFT TURN" after executing every "MOVE FORWARD" instruction, then identify the correct pair of the destination cell coordinates and direction after the drone executes three consecutive "MOVE FORWARD" instructions. Assume that the drone develops the fault when it is at cell coordinates (1, 5) facing North?
- (1) (2, 4) and North (2) (2, 2) and South (3) (1, 4) and East
- (4) (1, 3) and West (5) (3, 2) and West



Directions for questions 29 – 30: Read the following information carefully to answer the given questions. The tic-tac-toe game is played between "You" and your "Opponent" using the following set of instructions. The game stops when it is won by a player or the algorithm gets completed, whichever is earlier. A row means a vertical straight path or a horizontal straight path or a diagonal path.

The game is won when any player completes a row with the same set of marks.

(X or O)

Set of Instructions:

1. Occupy one of the corners.
2. Let the opponent take his turn.
3. Choose any unoccupied corner.
4. Let the opponent take his turn.
5. If you have 2 marks in a row with an empty space to make 3 in a row, mark the empty space. Otherwise, if the opponent has 2 marks in a row with an empty space to make 3 in a row, block the opponent. Otherwise, choose an open corner.
6. Let the opponent take his turn.
7. If you have 2 marks in a row with an empty space to make 3 in a row, mark the empty space. Otherwise, if the opponent has 2 marks in a row with an empty space to make 3 in a row, block the opponent. Otherwise, choose an empty spot.

29. Which of the following cannot be the result of the above set of instructions in a game of tic-tac-toe if you start the game with 'X'?

(1)

X	O	X
X	O	
O	X	

(2)

X		O
O	X	X
O		O

X	X	X ⁽³⁾
	O	O

(4)

X	O	X
	O	X
	X	O

(5)

X		O
X	O	
X	O	

30. Which of the following can be an outcome(s) for any of the players who initiate the game of tic-tac-toe?

1. Win for the player who initiates the game
2. Loss for the player who initiates the game
3. A draw for both the players

- (1) Only 1 (2) Only 2 (3) Both 1 and 3 (4) Both 2 and 3
(5) None of these



SECTION – 2: ENGLISH

Directions for questions 31 – 33: Read the following passage and answer the questions.

Corporate Social Responsibility (CSR) has become a critical aspect of contemporary business practices. The concept emphasises that companies should not only focus on maximising profits but also address the social, economic, and environmental issues that affect their stakeholders. In recent years, there has been a growing interest in CSR among businesses of all sizes. While some view CSR initiatives as a public relations tool, others believe they can provide tangible benefits to both the company and society.

Proponents of CSR argue that these initiatives can help companies build brand loyalty and enhance their reputation, thereby contributing to increased sales and revenue. For example, a company that engages in sustainable practices or supports social causes may be seen as socially responsible and ethical by its customers. This perception can translate into increased customer loyalty and a willingness to pay a premium for products or services from such a company.

Additionally, CSR initiatives can also help companies attract and retain employees, particularly younger generations who prioritise social and environmental responsibility. Employees are more likely to remain loyal to an organisation that demonstrates a commitment to social responsibility and ethical practices. Furthermore, CSR can create a positive workplace culture, boost employee morale and engagement, and attract top talent, resulting in a more productive and effective workforce.

Companies that prioritise CSR may also enjoy long-term financial benefits. Studies have shown that companies with strong CSR reputations tend to outperform their peers in terms of financial performance and long-term sustainability. Such companies often have a lower cost of capital, as investors view them as less risky investments. These companies are also more likely to be supported by governments, regulators, and other stakeholders, providing a favourable operating environment and reducing the potential for reputational risks.

Critics of CSR argue that a company's primary responsibility is to maximise profits for its shareholders, and that any efforts to prioritise social and environmental concerns are ultimately detrimental to the company's bottom line. They contend that the costs of CSR initiatives are often too high and detract from a company's core mission of generating profits.

In conclusion, CSR initiatives have become essential for companies that want to remain competitive and sustainable in today's business environment. These initiatives can help companies build brand loyalty, attract and retain employees, reduce risks, and enhance their financial performance in the long run.

31. Which of the following is/are potential advantage(s) that companies may gain from participating in CSR initiatives, as per the given text?

- (1) Companies that prioritise CSR initiatives can create a stronger bond with their clientele, which can result in increased sales and revenue.
- (2) CSR initiatives can overshadow the involvement of the company in any sociopolitical scandals, essentially compensating for the same.



- (3) CSR initiatives may help companies bypass government regulations and requirements, making it easier for them to operate.
- (4) Both (1) and (2)
- (5) All (1), (2) and (3)

32. 'CSR initiatives can also help companies attract and retain employees, particularly younger generations who prioritise social and environmental responsibility.' In elaboration of this sentence, which of these options follows?

- (1) Research shows that younger generations prioritise financial stability more than previous generations, resulting in a collective preference towards well-paying jobs.
- (2) Companies that prioritise CSR initiatives are likely to attract young employees, irrespective of their level of skill, who understand their environmental responsibility.
- (3) In today's business environment, employees are not just looking for a job that pays well, but they also want to work for a company that aligns with their values and beliefs.
- (4) CSR initiatives only provide benefits to the company itself and do not result in any significant advantage for employees, especially younger generations.
- (5) None of these

33. What is the argument of the critics of CSR?

- (1) That emphasis on CSR makes companies divert from their original focus
- (2) That corporate and social realms should be kept separate to ensure sustenance of the society
- (3) That implementing sustainable practices can lead to financial loss for certain markets
- (4) That CSR initiatives do not help companies in enhancing their brand recognition
- (5) None of these

Directions for questions 34 – 36: Read the following passage and answer the questions.

As an industry expert, I have seen firsthand the benefits and risks of technological development. On one hand, technology has revolutionised various industries, leading to increased efficiency, productivity, and innovation. On the other hand, there are concerns about the potential negative impact of technology on society, including privacy violations, job displacement, and widening inequality.

One of the biggest benefits of technology is the increased efficiency it brings to various industries. With the use of automation and machine learning, businesses can streamline their operations and reduce costs. This allows for greater profitability and competitiveness in the marketplace. Additionally, technology has made it easier for people to access information, connect with each other, and engage in commerce across borders.

However, technological advancement has also brought with it significant risks. One of the main concerns is privacy violations, as companies collect and use personal data without individuals' consent. This has led to debates around data protection and regulation, as well as concerns about surveillance and government overreach. Another risk is the potential for job displacement as automation and artificial intelligence take over tasks previously performed by humans. This can exacerbate inequality and leave many workers without viable employment opportunities.



Moreover, technological advancement can also exacerbate existing societal inequalities, as those who are already advantaged have greater access to resources and opportunities. This includes access to education and training programmes that are necessary to take advantage of technological advancements. This digital divide can lead to further disparities in income and wealth distribution.

In addition to the risks already mentioned, technological development has also raised concerns about the impact on mental health and social well-being. The rise of social media and digital communication has brought about a new form of cyber-addiction, cyberbullying, and online harassment. The constant barrage of information and stimuli can lead to increased stress and anxiety, affecting individuals' mental health and overall well-being.

Furthermore, technological advancement can have significant environmental consequences. The production and disposal of electronic devices and their components contribute to pollution and resource depletion. The rapid technological innovation has also led to a culture of planned obsolescence, where products are intentionally designed to have a short lifespan. This leads to increased waste and environmental degradation.

As society becomes more reliant on technology, it is essential to address these risks and challenges. This requires a multidisciplinary approach that involves not only industry experts but also policymakers, regulators, and civil society organisations. It is crucial to prioritise ethical considerations and ensure that technological development is aligned with societal values and goals. By doing so, we can maximise the benefits of technology while minimising its potential negative impact on society.

34. According to the passage, what is the potential downside to companies relying heavily on technology?
- (1) Potential for technological malfunctions or cyber-attacks leading to data breaches
 - (2) Unauthorised data collection and use leading to violation of individual privacy
 - (3) Over-reliance on technology leading to reduced critical thinking skills in employees
 - (4) Increased reliance on technology leading to decreased social interaction among peers
 - (5) None of these
35. According to the passage, how has technological development impacted the education sector?
- (1) It has increased access to educational resources and opportunities.
 - (2) It has resulted in decreased engagement and motivation among students.
 - (3) It has led to a shift towards online learning and distance education.
 - (4) It has created new challenges and barriers for underprivileged students.
 - (5) Both (1) and (3)
36. Which of the following categories best describes this piece of writing?
- (1) A chronological essay about technological development
 - (2) A comparison of conditions before and after technological development
 - (3) A composition oriented towards future aspirations regarding technology
 - (4) An analysis and critique of technology's influences
 - (5) Either (2) or (3)



Directions for questions 37 – 40: Read the following passage and answer the questions.

Calcutta was the seat of the government of Colonial India until 1911 and the commercial capital of the British Indian Empire down to the First World War. Once Bengal became the centre of English colonialism in the subcontinent, Calcutta and its hinterland rapidly emerged as the nucleus of British economic interests in the country. When the Raj sought to impart western education, Calcutta was the experimental site. The new western educated middle class rapidly made Calcutta the cultural capital of the subcontinent.

In the political realm, the early years of the 1910s were of momentous significance. When Lord Curzon, the Viceroy of India, announced the controversial partition of Bengal in 1905, the population of Calcutta spearheaded the Swadeshi Movement, renouncing British products for Indian goods. Other segments of the city soon joined the movement; in 1911, the goal of the movement was realised when the British government announced the annulment of Bengal's division. On the chilly but clear morning of December 12, 1911, Delhiites made a beeline for a maidan near Burari to witness the biggest, most over the top theatrics of the British Raj – the Delhi Durbar. The old city of Delhi, which had been the capital of Mughal India, was selected to accommodate the British.

The transfer of capital from Calcutta to Delhi indicated the British determination to evade future challenges to its authority. Although the move to Delhi was to take the wind out of the sails of anti – colonial Bengali politics, there was little doubt that Calcutta's status was much reduced by this move.

The decision to transfer the capital and abandon Calcutta was taken hurriedly. Lord Hardinge, the Viceroy of India, pointed out the pressing reasons to move away from Calcutta, which for 150 years had served as the British capital in India. The India Council's Act of 1909, a legislation passed by Britain's parliament, had allowed Indians to stand for legislative council position for the first time. For years, the British had ruled from Calcutta, which the East India Company, in the 18th century, had developed into a colonial city. Hardinge argued that the rising importance of the elected legislative bodies meant that Britain needed to find a more centrally located capital. The viceroy alluded to the burgeoning opposition to the British rule in Calcutta that was making it less than a hospitable home. The shift from Calcutta to Delhi symbolised British efforts to consolidate authority, transforming India's political and cultural landscape while altering power dynamics.

37. Which of the following statements accurately reflects the author's view of Calcutta before 1911, based on the given text?

- (1) It served as the administrative and trading hub of the British Indian Empire until the onset of World War I.
- (2) It was primarily known for its cultural significance and artistic achievements, despite the fact that it had limited access to resources and infrastructure.
- (3) It was a minor player in British India, overshadowed by other cities such as Bombay and Madras, and played little role in shaping the colonial government.
- (4) It was a centre of political power for Indian nationalists who sought to resist British colonialism and promote indigenous culture and traditions.
- (5) None of these



38. 'On the chilly but clear morning of December 12, 1911, Delhiites made a beeline for a maidan near Burari to witness the biggest, most over the top theatrics of the British Raj — the Delhi Durbar.' Which of the following best describes the author's view of the Delhi Durbar as expressed in the given sentence?

- (1) The author is admiring the grandeur and spectacle of the Delhi Durbar.
- (2) The author is criticising the excessive and showy nature of the Delhi Durbar.
- (3) The author is indifferent towards the Delhi Durbar and is only describing it factually.
- (4) The author is sceptical of the intentions and effects of the Delhi Durbar.
- (5) None of these

39. What was the main reason Lord Hardinge argued for the transfer of the capital to Delhi?

- (1) To contain the rising popularity of the elected legislatures of Indian origin
- (2) To weaken the nationalist sentiment in Calcutta and promote western education
- (3) To establish a new commercial hub and promote economic growth in India
- (4) To develop a new education system and promote modernisation in India
- (5) None of these

40. Which of the following statements regarding the Swadeshi Movement is/are correct?

- (1) The movement lasted for the better part of the 20th century.
- (2) The boycotters protested against the British attempt to stifle the nationalist sentiment.
- (3) Many British personalities allied themselves with the protesters against the British.
- (4) All of these
- (5) None of these

Directions: Read the following text and answer the question.

Human-elephant conflict is a complex problem which often pits local people's livelihoods and safety against the conservation of elephants. _____, there is a misperception that elephants live in wild places without people, and that conflict occurs only because people have encroached on elephant habitat. In reality, elephants are not confined to protected areas but live in shared landscapes, where there are no fences to separate people and elephants.

41. Which choice completes the text with the most logical transition?

- (1) Unfortunately
- (2) Additionally
- (3) Indeed
- (4) Similarly
- (5) None of these

Directions: In the following question, two statements numbered I and II are given. There may be cause and effect relationship between the two statements. These two statements may be the effect of the same cause or independent causes. These statements may be independent causes without having any relationship. Read both the statements in the question and mark your answer accordingly.

42. **Statements:**

- I. In India, suspects can be arrested and held for long periods before their case is decided by the courts.
- II. There is a very real possibility of long years of incarceration for persons who will eventually be proven to be innocent.



- (1) Statement I is the cause and statement II is its effect.
- (2) Statement II is the cause and statement I is its effect.
- (3) Both statements I and II are independent causes.
- (4) Both statements I and II are effects of independent causes.
- (5) Both statements I and II are effects of some common causes.

Directions: In the following question, four statements have been provided. These statements form a coherent paragraph when properly arranged. Select the alternative representing the proper and logical sequencing of these statements.

43. A. Whenever an insect species vanishes, or a language loses its last native speaker, the biosphere loses options.
B. Both as regards biological and cultural diversity, the threats are now massive and ominous.
C. Since the incipient environmentalist movement and the parallel concern among early anthropologists to document 'vanishing cultures', the visibility of human footprints and global cultural homogenisation have accelerated dramatically.
D. The main causes are the expansion of state and market forces, and the outcome can be described, in both cases, as a loss of flexibility.
- (1) ACBD (2) BDCA (3) DBAC (4) CBDA
(5) ABCD

Directions: Read the following text and answer the question.

For all we've learned about the sun, our home star remains shrouded in mystery. Now, after seven years of relative calm, the sun is set to become more _____ – and a fleet of sun-gazing spacecraft is ready to watch as it awakens. Those spacecraft are offering scientists an unprecedented chance to study our stormy star and the ways it can affect our cosmic neighbourhood.

44. Which choice completes the text with the most logical and precise word or phrase?
- (1) accessible (2) temperamental (3) mysterious
(4) transient (5) None of these

Directions: The following question consists of a pair of words, which have a certain relationship with each other. Select the alternative which bears the same relationship as the original pair of words does.

45. Ephemeral : Permanence
- (1) Commendable : Deprivation (2) Sordid : Intermittence (3) Spurious : Emulation
(4) Erratic : Predictability (5) Hacky : Parish

Directions: Read the following text and answer the question.

Revolutions broke out across Europe in the spring of 1848, but the French tumult was especially consequential. At the end of February, after bitter fighting in the streets, _____. The resulting power vacuum was filled by a provisional government assembled in great haste from a list drawn up by newspaper



editors and acclaimed by delirious crowds. A new electoral law replaced the tiny July Monarchy franchise with something approximating universal male suffrage.

46. Which choice completes the text so that it conforms to the conventions of Standard English?

- (1) the July Revolution of 1830 was benefitted by King Louis Philippe who fled Paris
- (2) the July Revolution of 1830 had benefitted King Louis Philippe who fled Paris
- (3) King Louis Philippe, the beneficiary of the July Revolution of 1830, fled Paris
- (4) King Louis Philippe's fleeing from Paris benefitted the July Revolution of 1830
- (5) None of these

Directions: Some alternative summaries are given below the text. Choose the option that best captures the essence of the text.

47. In a highly competitive, fast paced, globalised environment, organisational leaders need to judge others quickly. Across many types of decisions - whom to hire, which individuals will form the most innovative and effective teams, the best negotiation strategy to pursue – accurate judgments about others represent a key component for making good decisions.

When it comes to judging others, as much as we would like to believe that we assess each individual on his or her own idiosyncratic merits and flaws, we often do not have the time or luxury to exert the required cognitive resources. As a result, we make broad judgments that are less nuanced and less accurate.

- (1) Fast paced globalised environments cause leaders to err when making informed decisions about allocating the best people to tasks.
- (2) Many organisational decisions are erroneous as they are hasty, regardless of how well thought out and judiciously they may have been taken.
- (3) Good decision making is liable to suffer in an organisational environment hard pressed for time.
- (4) Matching people suitably to tasks is a vital component of good decision making, however fast paced the decision making environment be.
- (5) None of these

Directions: Read the following text and answer the question.

Studies show that forty percent of all students aged seventeen to twenty-five are residing outside their home districts in any given month. On the basis of these studies, a researcher concluded that sixty percent of all students aged seventeen to twenty-five are residents of their home districts year-round.

48. Which of the following, if true, would most seriously weaken the researcher's conclusion?

- (1) More students are residing outside their home districts today than during any other period since the nineteen fifties.
- (2) Many students, both males and females, change their residential status frequently.
- (3) Although only a small sample of the total student population is surveyed each month, these samples have been found to be a reliable indicator of monthly student residential status.
- (4) Surveys show that more students than ever before consider moving out of their home districts for education, which is an important priority.



- (5) Students who are residing outside their home districts have more independence than those residing in their home districts.

Directions: In the following question, two statements A and B are given. These statements may be either independent causes or may be effects of independent causes or of a common cause. One of these statements may be the effect of the other statement. Read both the statements and decide which of the following answer choices correctly depicts the relationship between the two statements.

Mark answer (1) if statement A is the cause and statement B is its effect.

Mark answer (2) if statement B is the cause and statement A is its effect.

Mark answer (3) if both statements A and B are independent causes.

Mark answer (4) if both statements A and B are effects of independent causes.

Mark answer (5) if both the statements A and B are effects of some common cause.

49. A. India needs to break the nexus between politics and real estate.
B. Real estate is a business where the builder depends on the government for the basic raw material - land.
- (1) (1) (2) (2) (3) (3) (4) (4) (5) (5)

Directions: Read the following text and answer the question.

To help determine predictors of success or failure in elite military training, researchers monitored the physical and psychological activity of Marines and sailors enrolled in a specialised training course. A total of 121 trainees participated. Only slightly more than half (64) successfully completed the course. Researchers found there was no correlation between course completion and physical performance, such as hikes or aquatic training. Physical markers such as heart rate or sleep status also did not play a role. Rather, the biggest determinant was mental. Trainees who identified themselves as extroverted and had a confident attitude were most likely to complete the course. These findings are novel because they identify traits not typically associated with military performance, showing that _____

50. Which choice most logically completes the text?
- (1) psychological factors matter more than physical performance outcomes.
(2) rigours of the training provide the reason why dropout rates are high.
(3) increase in emotional pain leads to decrease in confidence levels.
(4) baseline personality assessments are more important than military training.
(5) None of these



SECTION – 3: MATHEMATICS

51. If $7^m + 4^n = 13438$ and $\frac{1}{-(7)^{-(m+1)}} - \frac{1}{4^{(1-n)}} = 16526$, then the value of n equals
(1) 7 (2) 8 (3) 9 (4) 10
(5) None of these
52. Four friends, Ekam, Fazal, Gyan, and Hari, decide to cycle for 45 minutes on a circular cycling track in a park. Everyone cycle at different speeds. Ekam completes a lap exactly every 1 minute 10 seconds, Fazal completes a lap exactly every 30 seconds, Gyan completes a lap exactly every 40 seconds, and Hari completes a lap exactly every 1 minute. The friends start together at the starting point of the track exactly at 8:30 AM. How many laps will the friends have collectively completed when they next return to the starting point together?
(1) 60 (2) 75 (3) 80 (4) 82 (5) None of these
53. In a restaurant menu, price of every item is written as m% above the initial cost. After eating food, a customer gave m% extra of the price written as a tip to the restaurant owner; the restaurant owner made 125% profit. Find the value of m.
(1) 25 (2) 80 (3) 60 (4) 50 (5) 20
54. The number of positive integers which divide $(315) \times (96) \times (550)$ and are not divisible by 18 is _____.
(1) 432 (2) 288 (3) 256 (4) 172 (5) None of these
55. In a science fair, the scores of three participants, Rahul, Priya and Sameer, are such that Rahul's score is one-fourth of the sum of Priya's and Sameer's scores. After rechecking, each participant's score is increased by 15. The revised scores of Rahul, Priya and Sameer are in the ratio 3 : 4 : 5. Find the difference between Sameer's score and Priya's score.
(1) 15 (2) 20 (3) 25 (4) 10
(5) None of these
56. When working alone, Hema takes 75% more time to finish a job than Lily does. Lily and Hema complete the job in $56\frac{5}{7}$ days using the following manner: Lily works alone until one-third of the job is completed, then both Lily and Hema work together for 8 days. Finally, Hema works alone to finish $\left(\frac{2}{5}\right)^{th}$ of the job. How many days can it take for Hema to complete the entire job if she works alone?
(1) 36.5 days (2) 82.5 days (3) 40.5 days (4) 80 days
(5) None of these



57. Meena is a student at a music school. After each performance, she calculates her cumulative average score. In the first 6 performances, her average was 80. Scoring 101 points in Piano increased her average by 3. Scoring 99 points in Guitar further increased her average by 2. If Meena scores 130 in Singing, what will be her average score, given that Singing was her last performance?

- (1) 100 (2) 90 (3) 85 (4) 95 (5) None of these

58. In a school science fair, three students - Anand, Bipin, and Kiran - were awarded prize money in the ratio of 8 : 4 : 5 in the year 2018. In the year 2023, the ratio changed to 4 : 5 : 7. If Anand's prize money increased by 30% during the period 2018-2023, then what was the approximate percentage increase in Kiran's prize money during the same period?

- (1) 255% (2) 155% (3) 264% (4) 330% (5) None of these

59. Let $f(x)$ be a polynomial function such that when divided by $(x - 2)$, the remainder is 6, and when divided by $(x + 3)$, the remainder is -4. If the polynomial $f(x)$ is divided by $(x - 2)(x + 3)$, then the remainder will be

- (1) 2 (2) 4 (3) $2x + 2$ (4) $x + 4$ (5) None of these

60. In a football match, Team X scored a total of 53 points without missing a penalty kick. The points were accumulated from field goals, touchdowns, and successful extra-point conversions made by two players, Ajay and Vinay. The number of successful extra-point conversions was 12 times the points accumulated from touchdowns. Touchdowns points were 3 more than the number of field goals. If the ratio of the points scored by Ajay to that by Vinay was 3 : 5, then how many points did Ajay score?

- (1) 42 (2) 36 (3) 18 (4) 12 (5) 9

61. $F = 6P^2 - 3Q^2 + R + 0.5S$

It is given that P, Q, R, and S must be non-negative integers, and the following relationships must hold:

- (i) $2P + Q \leq 5$
(ii) $4P + 2Q + R \leq 10$
(iii) $2P + 2Q + 3S \leq 29$

Determine the value of S that maximises F, and also find the maximum value of F.

- (1) (8, 30) (2) (8, 82) (3) (12, 39) (4) (12, 93) (5) (21, 39)

62. A sum of Rs. 2,00,000 from a certain estate was divided among a spouse and three children. How much of the estate did the youngest child receive?

- (I) The spouse received $\frac{1}{2}$ of the sum from the estate, and the eldest child received $\frac{1}{4}$ of the remainder.
(II) Each of the two younger children received Rs. 12,500 more than the eldest child and Rs. 62,500 less than the spouse.

- (1) Statement (I) ALONE is sufficient, but statement (II) ALONE is not sufficient.
(2) Statement (II) ALONE is sufficient, but statement (I) ALONE is not sufficient.
(3) BOTH statements TOGETHER are sufficient, but NEITHER statement ALONE is sufficient.
(4) EACH statement ALONE is sufficient.
(5) Statements (I) and (II) TOGETHER are NOT sufficient.



63. Rohan buys two types of shirts, X and Y, for a total of Rs. 4,750. The number of type X shirts is 3 times the number of type Y shirts, and the cost per shirt of type X is 25% more than that of type Y. Rohan sells type X and type Y shirts with profits of 20% and 15%, respectively.

What is the overall profit in rupees?

- (1) 600 (2) 750 (3) 900 (4) 1,250 (5) 1,500

64. Together, Isha and Fatima earn as much money in a week as Garima does in four weeks. Together, Garima and Isha earn as much money in a week as Fatima does in two weeks. What is the ratio of the highest earner's weekly wages to the lowest earner's weekly wages?

- (1) 7 : 3 (2) 5 : 9 (3) 9 : 5 (4) 7 : 9 (5) None of these

65. Two cyclists, Amit and Bimal, participate in a 50 kilometre race. Amit starts the cycling at a constant speed of 12 kilometres per hour. After 30 minutes, Bimal starts cycling at a constant speed of 15 kilometres per hour. What is the distance from the starting point where Bimal will meet Amit?

- (1) 20 km (2) 30 km (3) 35 km (4) 40 km (5) None of these

66. In a 100-metre race, A beats B by 2 seconds and B beats C by 4 seconds. If A beats C by a distance 24 metres, how many seconds does A take to finish the race?

- (1) 19 seconds (2) 20 seconds (3) 25 seconds (4) 28 seconds (5) None of these

67. A company introduces a new promotional discount of 15% on its flagship product, resulting in a surge of 30% in the number of units sold. If the revenue generated by the company subsequently increases by $y\%$, then value of y is

- (1) 5% (2) 8.5% (3) 10.5% (4) 12% (5) None of these

68. A group of volunteers in a community organisation plans to prepare 2,800 gift baskets of chocolates to distribute to children in hospitals and community centres during a festive event. The initial plan for each volunteer is to make an equal number of gift baskets. However, on the day of preparation, fifteen volunteers are unable to attend, resulting in each remaining volunteer making 24 gift baskets more than originally decided. What is the new number of gift baskets made by each remaining volunteer?

- (1) 90 (2) 80 (3) 70 (4) 60 (5) 65

69. At a certain shop, the cost of a product is 'C' rupees. On the first day, due to an increase from the supplier, it goes up by ' $x\%$ ' and then it decreases by ' $y\%$ ' as a part of a promotional offer. On the second day, once again, it increases by ' $x\%$ ' due to another increase from the supplier, and then decreases by ' $y\%$ ' due to another promotional offer. As a result, the product's cost returns to 'C' rupees. If ' y ' is 25, what will be the cost of the product on the first day after the supplier increases the price?

- (1) $33.33C$ (2) $133.33C$ (3) $166.66C$ (4) $66.66C$ (5) $101.33C$

70. What is the approximate monthly instalment amount, rounded to the nearest rupee, for a car priced at Rs. 3,00,000 if you choose the alternative payment plan with an initial down payment of Rs. 75,000 and an 8% annual interest rate, distributing the remaining cost over four equal monthly payments?

- (1) Rs. 55,250 (2) Rs. 57,180 (3) Rs. 52,770 (4) Rs. 57,750 (5) Rs. 58,250



71. Among the cohort of 130 students who participated in the science fair, 60 exhibited their projects in the field of Biology (B), 80 of their peers showcased projects in Chemistry (C), and a total of 70 students took part in presenting projects related to Physics (P). Furthermore, it has been ascertained that at most 30 students concurrently displayed projects in both Biology and Chemistry, at most 40 students simultaneously presented projects in both Chemistry and Physics, and at most 20 students concurrently exhibited projects in both Biology and Physics. In light of this information, what is the maximum number of students who could have feasibly presented projects encompassing all three subjects: Biology, Chemistry, and Physics?

- (1) 8 (2) 10 (3) 15 (4) 18 (5) 20

72. Given the function $f(x - y) = \frac{f(x)}{f(y)}$, where x and y are positive integers and $f(1) = 3$, determine the value of r such that: $f(5) : f(a) = 1 : r$. This condition is based on the equation: $f(a - 1) + f(a - 2) +$

$$f(a - 3) + \dots + f(a - n) = \frac{(3^n - 1)}{18}.$$

(1) 7 (2) 8 (3) 9 (4) 11 (5) None of the above

73. In a basket of fruits, there are 21 apples, out of which exactly 12 are bruised. Each day, one apple is randomly chosen from the basket. What is the probability that after 5 days, there are exactly 8 apples in the basket that are not bruised?

- (1) $\frac{99}{2,261}$ (2) $\frac{405}{2,261}$ (3) $\frac{495}{2,261}$ (4) $\frac{549}{2,261}$ (5) $\frac{945}{2,261}$

74. The 10th, 15th, and 25th terms of an arithmetic progression form three consecutive terms of a geometric progression. If the next term of the geometric progression is the n^{th} term of the arithmetic progression, then what is the value of 'n'?

- (1) 42 (2) 45 (3) 47 (4) 48 (5) None of these

75. If $\int e^x \left\{ \alpha \log(e^{2 \log x} + 1) + \frac{\beta x}{e^{2 \log x} + 1} \right\} dx = \frac{\beta}{2} e^x \log(x^2 + 1)$, then

- (1) $\alpha = \frac{\beta}{2}$ (2) $\beta = 1, \alpha = 2$ (3) $\beta = 1, \alpha = 1$ (4) $\alpha = \beta$ (5) None of these

76. Let $f: [0, 1] \rightarrow \mathbb{R}$ be the function defined as $f(x) = 2^{\sin x} + 2^{\sin(1-x)}$, where $m = \sin^2\left(\frac{\pi}{6}\right) + \sin^4\left(\frac{\pi}{6}\right) + \sin^6\left(\frac{\pi}{6}\right) + \dots + \infty$. Then which of the following statements is not true?

- (1) The minimum value of the function is $2^{\frac{7}{6}}$.
(2) The maximum value of the function is $1 + 2^{\frac{1}{3}}$.
(3) The maximum value of the function is attained at two different points.



- (4) The minimum value of the function is attained at two different points.
(5) The maximum value of the function is attained at $x = 0$.

77. If $af(x) + bf\left(\frac{1}{x}\right) = x^2 + \frac{1}{x^2}$, where $a \neq b$, then $f'(2) =$ _____.

- (1) $\frac{15}{4(a+b)}$ (2) $\frac{3}{4(a+b)}$ (3) $\frac{7}{4(a+b)}$ (4) $\frac{5}{4(a+b)}$ (5) $\frac{11}{4(a+b)}$

78. If P and Q are non-singular matrices such that $P - Q$ is also a non-singular matrix and $PQ = QP$, then $2P - Q - P(P - Q)^{-1}P + Q(P - Q)^{-1}Q$ equals

- (1) P (2) Q (3) $P - Q$ (4) $P - 2Q$ (5) None of these

79. Consider the matrix $A = \begin{bmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 - 1 & b^3 - 1 & c^3 - 1 \end{bmatrix}$, where a, b and c are real numbers, such that $|A| = 0$. Then the value of abc will be ____.

- (1) -2 (2) -1 (3) 0 (4) 1 (5) 2

80. Consider sets $M = \{1, 2, 3, 4\}$ and $N = \{p, q\}$. Assuming all the relations from set M to set N are equally likely, then what is the probability that a relation from M to N is not a function?

- (1) $\frac{1}{16}$ (2) $\frac{5}{16}$ (3) $\frac{7}{16}$ (4) $\frac{15}{16}$ (5) None of these



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NSAT Phase-1 Solutions (1)

ANSWER KEY

- | | | | | | | | | | | | | | |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1. | (2) | 2. | (3) | 3. | (1) | 4. | (1) | 5. | (4) | 6. | (5) | 7. | (3) |
| | 8. | | (4) | | | | | | | | | | |
| 9. | (2) | 10. | (2) | 11. | (2) | 12. | (1) | 13. | (1) | 14. | (2) | 15. | (3) |
| | 16. | | (4) | | | | | | | | | | |
| 17. | (1) | 18. | (3) | 19. | (2) | 20. | (3) | 21. | (2) | 22. | (1) | 23. | (4) |
| | 24. | | (5) | | | | | | | | | | |
| 25. | (5) | 26. | (4) | 27. | (4) | 28. | (3) | 29. | (2) | 30. | (3) | 31. | (1) |
| | 32. | | (3) | | | | | | | | | | |
| 33. | (1) | 34. | (2) | 35. | (4) | 36. | (4) | 37. | (1) | 38. | (2) | 39. | (1) |
| | 40. | | (2) | | | | | | | | | | |
| 41. | (1) | 42. | (5) | 43. | (4) | 44. | (2) | 45. | (4) | 46. | (3) | 47. | (3) |
| | 48. | | (2) | | | | | | | | | | |
| 49. | (2) | 50. | (1) | 51. | (1) | 52. | (2) | 53. | (4) | 54. | (3) | 55. | (4) |
| | 56. | | (2) | | | | | | | | | | |
| 57. | (2) | 58. | (3) | 59. | (3) | 60. | (3) | 61. | (1) | 62. | (2) | 63. | (3) |
| | 64. | | (1) | | | | | | | | | | |
| 65. | (2) | 66. | (1) | 67. | (3) | 68. | (2) | 69. | (2) | 70. | (2) | 71. | (2) |
| | 72. | | (3) | | | | | | | | | | |
| 73. | (3) | 74. | (2) | 75. | (1) | 76. | (4) | 77. | (1) | 78. | (4) | 79. | (4) |
| | 80. | | (4) | | | | | | | | | | |

EXPLANATIONS

1. **Answer:** (2)

Animal	Name	Favourite Fruit	
Fox		Terry	From (1)
	Dikky	Merry	From (2)
Hare	Nikky		From (3)
Monkey	Chikky		From (3)
Elephant		Cherry	From (4)

From (4), the Monkey doesn't like Ferry. So, the Hare has to like Ferry and the Monkey likes Berry.

From (1), the Fox's name is not Rikky, so it is Mikky and the Elephant is Rikky.

So, the table is as follows:

Animal	Name	Favourite Fruit
Fox	Mikky	Terry
Bear	Dikky	Merry
Hare	Nikky	Ferry
Monkey	Chikky	Berry
Elephant	Rikky	Cherry



2. **Answer:** (3)

Animal	Name	Favourite Fruit	
Fox		Terry	From (1)
	Dikky	Merry	From (2)
Hare	Nikky		From (3)
Monkey	Chikky		From (3)
Elephant		Cherry	From (4)

From (4), the Monkey doesn't like Ferry. So, the Hare has to like Ferry and the Monkey likes Berry.

From (1), the Fox's name is not Rikky, so it is Mikky and the Elephant is Rikky.

So, the table is as follows:

Animal	Name	Favourite Fruit
Fox	Mikky	Terry
Bear	Dikky	Merry
Hare	Nikky	Ferry
Monkey	Chikky	Berry
Elephant	Rikky	Cherry

3. **Answer:** (1)

Animal	Name	Favourite Fruit	
Fox		Terry	From (1)
	Dikky	Merry	From (2)
Hare	Nikky		From (3)
Monkey	Chikky		From (3)
Elephant		Cherry	From (4)

From (4), the Monkey doesn't like Ferry. So, the Hare has to like Ferry and the Monkey likes Berry.

From (1), the Fox's name is not Rikky, so it is Mikky and the Elephant is Rikky.

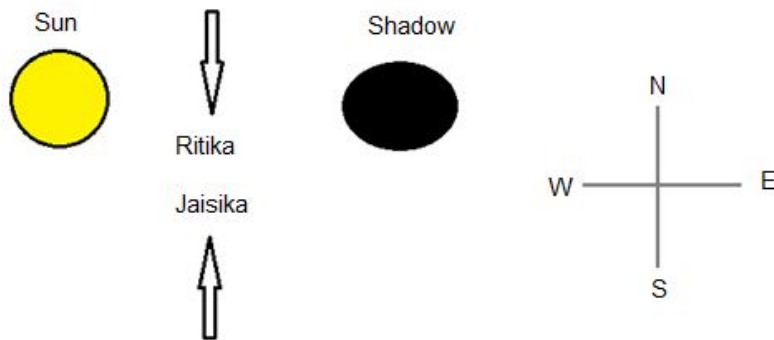
So, the table is as follows:

Animal	Name	Favourite Fruit
Fox	Mikky	Terry
Bear	Dikky	Merry
Hare	Nikky	Ferry
Monkey	Chikky	Berry
Elephant	Rikky	Cherry



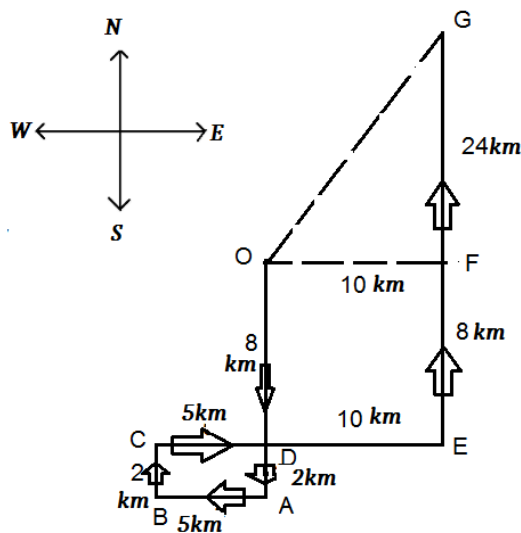
4. **Answer:** (1)

At the time of evening, the sun is in west direction. So, the shadow will form in east direction. If the shadow of Jaisika was exactly to the left of Ritika, then Jaisika was facing north direction.



After taking a turn of 90° clockwise, she would be facing east direction.

5. **Answer:** (4)



In triangle OGF, applying Pythagoras theorem,

$$\Rightarrow OG^2 = OF^2 + GF^2$$

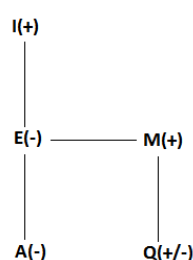
$$\Rightarrow OG^2 = 10^2 + 24^2$$

$$\Rightarrow OG^2 = 100 + 576 = 676$$

$$\Rightarrow OG = 26 \text{ km}$$

6. **Answer:** (5)

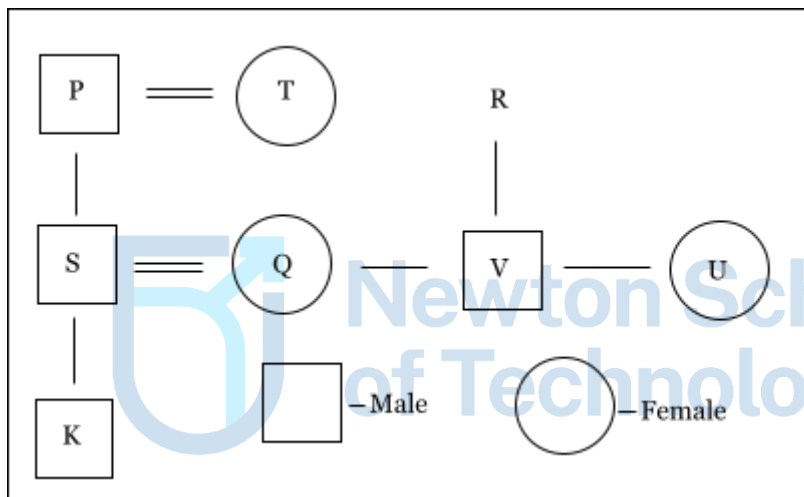
According to the given information, A is the daughter of E, who is the daughter of I. I is the father of M, who is the father of Q. So, all of the relations given in the options are incorrect.





7. **Answer:** (3)

According to the given information, the following diagram is obtained:

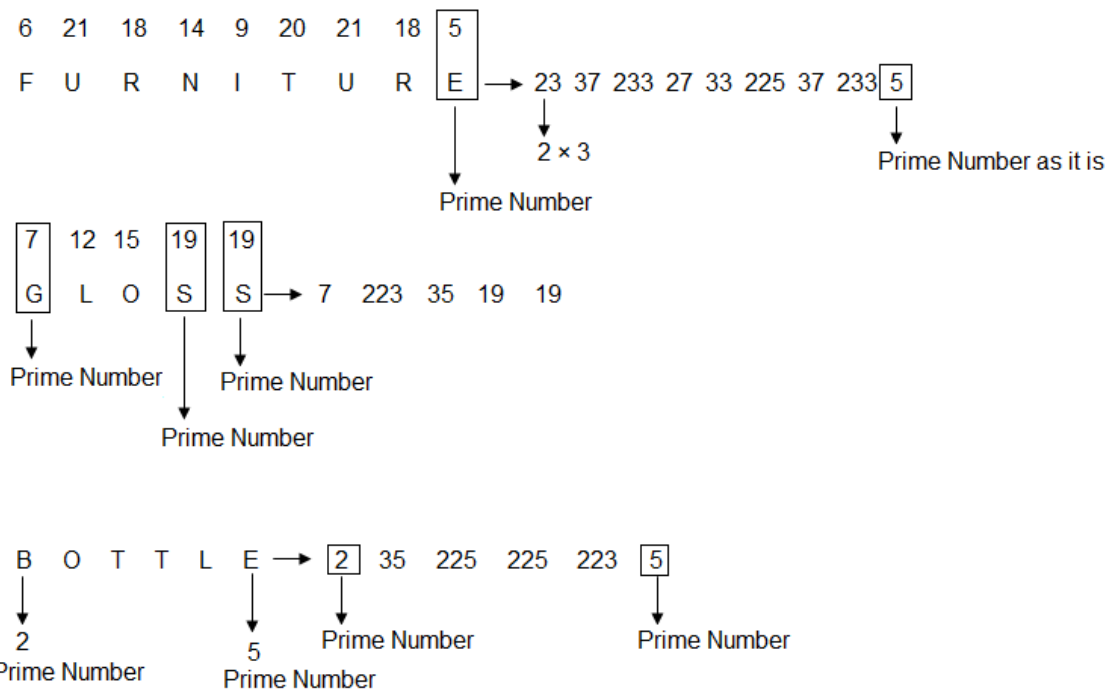


So, V is K's uncle.

8. **Answer:** (4)

Code is in the form of prime factorisation.

Prime number is as it is.



Hence, the code is [2352252252235].

9. **Answer:** (2)

The pattern followed here is:

Letter - Last letter of the word

Number - Alphabetical position of the first letter of the word

The code for 'believe' will be '2E'.



10. **Answer: (2)**

The pattern followed here is:

Letter - Last letter of the word

Number - Alphabetical position of the first letter of the word

The word for '20W' would be 'tomorrow'.

So, 'Hope for tomorrow' will be coded as '8E 6R 20W'.

11. **Answer: (2)**

No. of average students in 2012 = $120 + 100 + 98 = 318$

No. of above average students in 2016 = $60 + 28 + 78 + 18 + 88 + 28 = 300$

Required ratio = $318 : 300 = 53 : 50$

12. **Answer: (1)**

Percentage of good grade students of secondary level:

$$2012: \frac{68}{100 + 68 + 24} \times 100 = 35.41$$

$$2013: \frac{75}{98 + 75 + 22} \times 100 = 38.46$$

$$2014: \frac{60}{105 + 60 + 25} \times 100 = 31.57$$

$$2015: \frac{70}{110 + 70 + 28} \times 100 = 33.65$$

$$2016: \frac{78}{125 + 78 + 18} \times 100 = 35.29$$

$$2017: \frac{56}{120 + 56 + 21} \times 100 = 28.42$$

Therefore, the percentage of good grade students of secondary level is highest in 2013.

13. **Answer: (1)**

In 2012, total student strength = $120 + 75 + 35 + 100 + 68 + 24 + 98 + 82 + 29 = 631$

In 2013, total student strength = $115 + 68 + 34 + 98 + 75 + 22 + 89 + 78 + 37 = 616$

In 2014, total student strength = $132 + 54 + 30 + 105 + 60 + 25 + 95 + 90 + 29 = 620$

In 2015, total student strength = $145 + 64 + 25 + 110 + 70 + 28 + 102 + 75 + 34 = 653$

In 2016, total student strength = $128 + 60 + 28 + 125 + 78 + 18 + 99 + 88 + 28 = 652$

In 2017, total student strength = $110 + 80 + 32 + 120 + 56 + 21 + 110 + 90 + 32 = 651$

So, the number of students was the highest in the year 2015.

14. **Answer: (2)**

$$\text{The number of people who preferred to travel by bike in 2011} = 5\% \text{ of } 12,000 = \frac{5}{100} \times 12,000 =$$



The number of people who preferred to travel by bus in 2017 = 25% of $40,000 = \frac{25}{100} \times 40,000$
= 10,000
Required ratio = 600 : 10,000
= 6 : 100
= 3 : 50

15. **Answer: (3)**

The total number of people who preferred to travel by bus in 2012 and 2015 together = 42% of 15,000

$$+ \quad 32\% \text{ of } 24,000 = \frac{42}{100} \times 15,000 + \frac{32}{100} \times 24,000 = 6,300 + 7,680 = 13,980$$

The total number of people who preferred to travel by metro in 2014 and 2017 together = 50% of

$$20,000 + 62\% \text{ of } 40,000 = \frac{50}{100} \times 20,000 + \frac{62}{100} \times 40,000 = 10,000 + 24,800 = 34,800$$

Required difference = $34,800 - 13,980 = 20,820$

16. **Answer: (4)**

The total number of people who preferred to travel by metro in 2017 = 62% of 40,000 = 24,800

The total number of people who preferred to travel by bus in 2015 = 32% of 24,000 = 7,680

Difference = $24,800 - 7,680 = 17,120$

$$\text{Required percentage} = \frac{17,120}{7,680} \times 100\%$$

$$= \frac{1,712}{768} \times 100\% \\ = 222.91\% \approx 223\%$$

17. **Answer: (1)**

The total number of male teachers from Rajasthan, Bihar and Gujarat together = $(20\% + 14\% + 14\%)$ of 3 lakh

$$= 48\% \text{ of } 3,00,000 = 1,44,000$$

The total number of female teachers from UP, Delhi and Punjab together = $(15\% + 25\% + 10\%)$ of 2.5 lakh

$$= 50\% \text{ of } 2,50,000 = 1,25,000$$

Thus, the required percentage = $(1,44,000 / 1,25,000) \times 100 = 115.2\%$

Answer option (1) is the correct answer.

18. **Answer: (3)**

The total number of teachers (male + female) from Delhi = 24% of 3 lakh + 25% of 2.5 lakh

$$= 24\% \text{ of } 3,00,000 + 25\% \text{ of } 2,50,000 = 72,000 + 62,500 = 1,34,500$$

The total number of teachers (male + female) from UP = 16% of 3 lakh + 15% of 2.5 lakh

$$= 16\% \text{ of } 3,00,000 + 15\% \text{ of } 2,50,000 = 48,000 + 37,500 = 85,500$$

The total number of teachers (male + female) from Bihar = 14% of 3 lakh + 16% of 2.5 lakh



$$= 14\% \text{ of } 3,00,000 + 16\% \text{ of } 2,50,000 = 42,000 + 40,000 = 82,000$$

The total number of teachers (male + female) from Punjab = 12% of 3 lakh + 10% of 2.5 lakh

$$= 12\% \text{ of } 3,00,000 + 10\% \text{ of } 2,50,000 = 36,000 + 25,000 = 61,000$$

$$\text{Thus, } A = 1,34,500 + 85,500 = 2,20,000$$

$$B = 82,000 + 61,000 = 1,43,000$$

$$\text{The value of } (A + B) = (2,20,000 + 1,43,000) = 3,63,000$$

This is the correct answer.

19. **Answer:** (2)

The total number of male teachers from Delhi, UP and Rajasthan together = (24% + 16% + 20%) of 3 lakh

$$= 60\% \text{ of } 3 \text{ lakh} = 60\% \text{ of } 3,00,000 = 1,80,000$$

The total number of female teachers from Bihar, Gujarat and Punjab together = (16% + 16% + 10%) of 2.5 lakh

$$= 42\% \text{ of } 2.5 \text{ lakh} = 42\% \text{ of } 2,50,000 = 1,05,000$$

$$\text{Thus, required difference} = 1,80,000 - 1,05,000$$

$$= 75,000 \text{ more}$$

20. **Answer:** (3)

From the above graph,

For 15 hours of the revision time period, marks obtained are shown in the vertical axes which is 35%.

As maximum marks of the exam = 400,

Possible marks obtained by the student must be nearly 35% of 400 = 140.

By options, 140 is the required marks obtained by the student.

21. **Answer:** (2)

As Abhay's date of birth is 30th June, 1993, his age meets the age criterion as laid down in criterion 1.

His education background and academic records meet criterion 2. His physical attributes and marital

status fulfil criteria 3 and 4. His AFCAT score of 75% fulfils criterion 5. Thus, he should be called for

the interview round. Hence, answer option (2) is correct.

22. **Answer:** (1)

Gurmeher meets all criteria except 1 and 2.3 as she is a B.E. in Computer Science, which is not one of the specified streams.

Pilot She gets an exemption for 1, as could not be working with Jet Airways without a valid Commercial Licence issued by DGCA. Her age limit is $28 < 29$ i.e. (24 + 5) years.

She may, also, be exempted from 2.3 if she fulfils criterion 5, i.e. a minimum AFCAT score of 60%.

There is no information given in the question regarding her score in AFCAT. Hence, data is insufficient.

23. **Answer:** (4)

Rajkumar fulfils all criteria, except 2.2 as he has scored 64% marks against a requirement of 65%. For

this, he may be granted an exemption if he meets criterion 5 (i.e. he scores a minimum of 60% marks



in AFCAT, consequent to which he would be allowed an exemption of 5% in his B.Sc. score) and is a sportsperson at the university level. Rajkumar fulfils criterion 5 as he has scored 72% in AFCAT. This means, his minimum score requirement in B.Sc. is $65\% - 5\% = 60\%$, which is less than his score of 64%. Also, since he was a captain in his university soccer team, he fulfils both the conditions needed to exempt him from criterion 2.2. His case needs to be forwarded to the selection committee.

24. **Answer:** (5)

As Swati's birthday is on the 31st of December, she is eligible to appear for the exam. Her education qualifications meet the criteria set in point 2. Her marital status meets criterion 4. Her AFCAT score also satisfies condition 5. Her physical characteristics satisfy conditions 3.1 and 3.2. However, no information has been given as to whether condition 3.3 can be met or not. Thus, the given data is not sufficient to answer the question. Hence, answer option 5 is correct.

25. **Answer:** (5)

The drone is initially at (2, 3) and facing North.

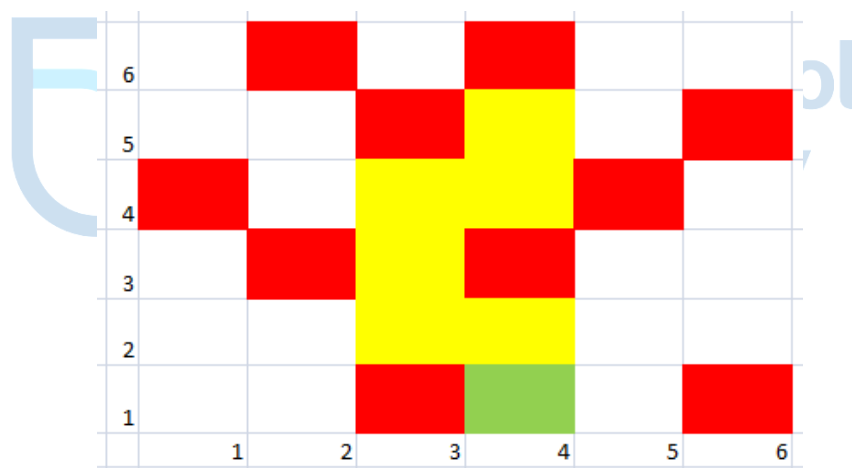
After moving forward, the drone will reach (3, 3).

After moving forward again, the drone will reach (4, 3).

Making a left turn again means the drone will change direction only, its position will remain in the same cell, i.e. (4, 3). Hence, the answer is 5. None of these



26. **Answer:** (4)

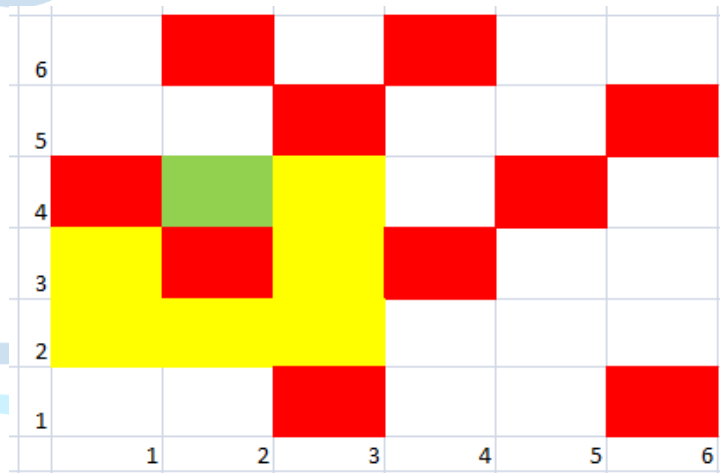


The complete set of instructions is as given below:



1. MOVE FORWARD
2. TAKE LEFT TURN
3. MOVE FORWARD
4. TAKE RIGHT TURN
5. MOVE FORWARD
6. MOVE FORWARD
7. TAKE RIGHT TURN
8. MOVE FORWARD
9. TAKE LEFT TURN
10. MOVE FORWARD

27. **Answer:** (4)



28. **Answer:** (3)

Upon executing the first "MOVE FORWARD" instruction, the drone reaches cell (2, 5) and then automatically turns left.

Upon executing the second "MOVE FORWARD" instruction, it reaches cell (2, 4). Again, it automatically turns left.

Upon executing the third "MOVE FORWARD" instruction, it reaches cell (1, 4). Again, it automatically takes a left turn. Now it faces East.

29. **Answer:** (2)

Since "You" start the game with the mark 'X', the mark must appear in at least 2 corner squares. Only in option 2, no two corners have the mark 'X'.

30. **Answer:** (3)

In the tic-tac-toe game where Player 1 initiates the game and follows the given set of instructions, the possible outcomes are:

Win for Player 1: This occurs if Player 1 successfully follows the instructions, which include occupying corners and blocking the opponent's moves. If Player 2 makes suboptimal moves, Player 1 can capitalize on these to win.

Draw: If both players play optimally, the game is likely to end in a draw. The instructions ensure that



Player 1 both attempts to win and blocks Player 2's winning moves, leading to a stalemate if Player 2 also plays strategically.

A loss for Player 1 is not a probable outcome when following these instructions and assuming optimal play from both players.

31. **Answer: (1)**

The correct answer is option 1 - Companies that prioritise CSR initiatives can create a stronger bond with their clientele, which can result in increased sales and revenue. The given text explains that companies that engage in CSR initiatives may build customer loyalty and enhance their reputation, leading to a willingness among customers to pay a premium for their products or services. This means that customers are more likely to remain loyal to a company that they perceive as socially responsible and ethical, which can result in increased sales and revenue for the company. Option 2 is incorrect because CSR initiatives cannot compensate for a company's involvement in sociopolitical scandals, as such scandals can have a significant negative impact on a company's reputation and brand image. Option 3 is also incorrect because CSR initiatives do not help companies bypass government regulations or requirements, but instead encourage companies to adhere to ethical and sustainable practices while complying with applicable laws and regulations.

32. **Answer: (3)**

The correct option that elaborates on the given sentence is Option 3: 'In today's business environment, employees are not just looking for a job that pays well, but they also want to work for a company that aligns with their values and beliefs.' This option reinforces the idea that younger generations prioritise social and environmental responsibility when it comes to choosing an employer. The option also emphasises the importance of aligning personal values and beliefs with the company's values, which is a critical factor in employee retention. Option 1 is incorrect because it talks about financial stability rather than social and environmental responsibility. Option 2 is incorrect because it makes a generalised statement that may not be true for all young employees. Option 4 is incorrect because it contradicts the argument presented in the original sentence.

33. **Answer: (1)**

Refer to, 'Critics of CSR argue that a company's primary responsibility is to maximise profits for its shareholders, and that any efforts to prioritise social and environmental concerns are ultimately detrimental to the company's bottom line.' Critics argue that implementing CSR initiatives can divert companies from their original focus, which is to generate profits for their shareholders. They contend that any resources spent on social or environmental concerns take away from the company's core mission and are ultimately detrimental to its financial performance.

34. **Answer: (2)**

The potential downside to companies relying heavily on technology, according to the passage, is unauthorised data collection and use, which can lead to violations of individual privacy. The passage mentions that companies collecting and using personal data without individuals' consent has led to debates around data protection and regulation, as well as concerns about surveillance and government overreach. Therefore, option (2) is the correct answer. Other options are not mentioned in the text.



35. **Answer:** (4)

The passage mentions that technological advancements can exacerbate existing societal inequalities, and one of the ways this happens is through unequal access to education and training programmes necessary to take advantage of these advancements. This leads to a digital divide that can result in further disparities in income and wealth distribution. In particular, underprivileged students may not have access to the technology or infrastructure necessary to participate in online learning and other digital educational resources, which can further disadvantage them in terms of educational opportunities and future job prospects.

36. **Answer:** (4)

The passage is a reflective piece that discusses the benefits and risks of technological development. It analyses the impact of technology on various aspects of society, such as privacy, employment, education, mental health, and the environment. The author offers a critical perspective on technology and suggests the need for a multidisciplinary approach to address its risks and challenges. Therefore, the best description of this piece of writing is "An analysis and critique of technology's influences".

37. **Answer:** (1)

The text mentions that Calcutta was the seat of the government of Colonial India until 1911 and the commercial capital of the British Indian Empire down to the First World War. It also states that Calcutta rapidly emerged as the nucleus of British economic interests in the country after Bengal became the centre of English colonialism in the subcontinent. Therefore, the correct answer is option 1.

38. **Answer:** (2)

Option 2, "The author is criticising the excessive and showy nature of the Delhi Durbar," is the best description of the author's view of the event. This can be inferred from the use of the phrase "biggest over the top theatrics" to describe the event, which implies that the author believes the Durbar was excessively showy and theatrical.

39. **Answer:** (1)

Lord Hardinge argued that the rising importance of the elected legislative bodies, which included Indian representatives for the first time, meant that Britain needed to find a more centrally located capital. Additionally, the burgeoning opposition to British rule in Calcutta was also a concern. Hardinge alluded to the fact that the British were ruling from Calcutta, the commercial hub of India, which the East India Company, in the 18th century, had developed into a colonial city. The rising popularity of the Indian-origin elected legislatures posed a challenge to the British authority, and they needed to find a more secure location to establish their rule.

40. **Answer:** (2)

The Swadeshi Movement was a socio-economic and political movement in India during the early 20th century, which aimed at achieving self-sufficiency and boycotting British goods. The movement was launched in response to the British partition of Bengal in 1905, which was seen as an attempt to weaken the nationalist sentiment in the region. The Swadeshi Movement encouraged Indians to boycott British goods and use Indian-made products instead.



41. **Answer:** (1)

The text appears to clear off a misconception that arises out of human-elephant conflict. The misconception is that it is caused because humans have encroached on elephant habitats. This is evident from the use of the words 'In reality' in the last sentence. In this sentence, though, the author laments the fact that there is a misconception. So, 'unfortunately' will be the correct option. Option 2 is incorrect because 'additionally' would suggest that there is an additional idea being conveyed by this sentence, which is not correct. 'Indeed' would suggest that this sentence elaborates the idea stated in the previous sentence, which is also incorrect. 'Similarly' is incorrect because this sentence does not indicate a similar idea as stated in the previous sentence.

42. **Answer:** (5)

Statement 'I' and 'II' are similar to each other, both the statements describes the possibility of suspect being arrested / incarcerated for long duration until their case is decided/until they are proven innocent or guilty. So, clearly Both statements I and II are effects of some common causes.

43. **Answer:** (4)

BDA form a sequence. 'The main causes ...' in D links with 'threats' mentioned in B. Thus, the main causes of 'threats' to cultural and biological diversity are the expansion of state and market forces. A logically follows D as it gives an example of 'loss of flexibility' described in D. Thus, 'loss of flexibility' in D links with 'loses options' in A. Only option 4 exhibits this sequence. Thus, CBDA is the correct sequence.

44. **Answer:** (2)

The text establishes in the same sentence that sun has remained calm for seven years after which it would naturally become more excited. This is even suggested later in the text by stating that "it awakens" or "our stormy stars" affecting "our cosmic neighbourhood". The correct word to describe this would be 'temperamental' or excited or unstable since it provides a contrast to 'calm' stated earlier in the sentence.

Option 1 ('accessible') cannot be used in the context of sun as the text does not mention anything about reaching the sun.

Option 3 ('mysterious') is incorrect as being temperamental strikes a contrast between periods of calm and outburst but mysterious refers to a quality which has already been mentioned in the first sentence.

Option 4 ('transient') which means temporary would be incorrect because the text does not suggest anything like that.

45. **Answer:** (4)

The given pair is a pair of antonyms. 'Ephemeral' means temporary and 'permanence' is its opposite. 'Erratic' means not regular in pattern or movement; unpredictable. 'Predictability' refers to consistent repetition of a state, course of action, behaviour. Thus, option 4 bears the same analogy as the original pair.

(1) 'Commendable' means something deserving praise. 'Deprivation' means lack or denial of something considered to be a necessity.

(2) 'Sordid' means dirty or filthy. 'Intermittence' refers to the quality of being intermittent (non-continuous).

(3) 'Spurious' means false or fake. 'Emulation' means matching or surpassing a person or achievement, typically by imitation.

(5) 'Hacky' means lacking originality. 'Parish' refers to a small administrative district typically having its own church and a priest. These are unrelated words.



46. **Answer:** (3)

Option 3 is the correct answer. The appositive "the beneficiary of the July Revolution of 1830" provides additional information about "King Louis Philippe" and should be marked off with a comma on both the sides. Options 1 and 2 are incorrect because they change the meaning. Option 4 is incorrect because using it would incorrectly modify "King Louis Philippe's fleeing from Paris with after bitter fighting in the streets".

47. **Answer:** (3)

The paragraph stresses "good decision making" the spirit of which is "the accuracy of the decision". Refer to the sentence that captures the essence of the paragraph - accurate judgments about others represent a key component for making good decisions.

Now when an organisation is short on time, the decision making process fails as the decisions lose their accuracy.

This has been captured only by option 3.

48. **Answer:** (2)

The conclusion that the researcher draws is that sixty percent students reside in their home districts year-round; it will be most seriously weakened if students keep changing their residential status frequently.

That way, the result of the studies would hold but the conclusion drawn by the author would not. Hence, option (2) is correct.

49. **Answer:** (2)

The cause of statement A is probably some problem or advantage that any party tends to take in the nexus. Statement B does not present that problem or advantage.

The cause of statement B is again the fact that government controls the land, which is not the same as statement A.

50. **Answer:** (1)

Option 1 is the correct answer. The findings are that, 'Physical markers such as heart rate or sleep status also did not play a role...Trainees who identified themselves as extroverted and had a confident attitude were most likely to complete the course'. That means the latter mattered more than the former.

Option 2 is incorrect. Rigours of the training could refer to any type of rigours, more likely physical.

That actually goes against the research findings.

Option 3 is incorrect. The text discusses confidence levels as determinants rather than as consequence.

Option 4 is incorrect. Comparison of rigorous military training with personality assessments is entirely misplaced in context.

51. **Answer:** (1)

$$\text{Given: } 7^m + 4^n = 13438 \dots\dots(1)$$

$$\text{And } \frac{1}{-(7)^{-(m+1)}} - \frac{1}{4^{(1-n)}} = 16526$$

$$\Rightarrow -7^{(m+1)} - 4^{(n-1)} = 16526 \dots\dots(2)$$

Multiplying equation (1) by 7 and adding in equation (2), we get



$$\Rightarrow 7(4^n) - \frac{4^n}{4} = 110592$$

$$\Rightarrow 4^n \left(\frac{27}{4} \right) = 110592$$

$$\text{So, } 4^n = 16384 = 4^7$$

We get $n = 7$.

52. **Answer:** (2)

Time taken by four friends to complete one lap:

Ekam takes 1 minute 10 seconds = 70 seconds.

Fazal takes 30 seconds, Gyan takes 40 seconds and Hari takes 1 minute = 60 seconds.

LCM of 70, 30, 40, 60 = 840

Number of laps by Ekam in 840 seconds = $840 \div 70 = 12$

Number of laps by Fazal in 840 seconds = $840 \div 30 = 28$

Number of laps by Gyan in 840 seconds = $840 \div 40 = 21$

Number of laps by Hari in 840 seconds = $840 \div 60 = 14$

Total number of laps completed = $12 + 28 + 21 + 14 = 75$

53. **Answer:** (4)

Let CP = 100

MP = $(100 + m)$

$$\text{SP} = (100 + m) \left(1 + \frac{1}{100} m \right)$$

$$\text{SP} - \text{CP} = 125$$

$$(100 + m) \left(1 + \frac{1}{100} m \right) - 100 = 125$$

$$100 + m + m + 0.01m^2 - 100 = 125$$

$$0.01m^2 + 2m - 125 = 0$$

$$m^2 + 200m - 12,500 = 0$$

$$m^2 + 250m - 50m - 12,500 = 0$$

$$m(m + 250) - 50(m + 250) = 0$$

$$(m + 250)(m - 50) = 0$$

$$m = 50, -250$$

$$m = 50 \text{ (because it can't be negative)}$$

54. **Answer:** (3)

First, let's find the prime factorisation of each number:

$$315 = 3^2 \times 5 \times 7$$

$$96 = 2^5 \times 3$$

$$550 = 2 \times 5^2 \times 11$$

$$(315) \times (96) \times (550) = 2^6 \times 3^3 \times 5^3 \times 7 \times 11$$

$$\text{Total number of factors} = (6 + 1)(3 + 1)(3 + 1)(1 + 1)(1 + 1) = 448$$



We need to exclude the factors that contribute to divisibility by $18 = 2 \times 3^2$

For the prime factor 2, the highest power that can be included is 2^5 .

For the prime factor 3, the highest power that can be included is 3^1 .

For the prime factor 5, the highest power that can be included is 5^3 .

For the prime factor 7, the highest power that can be included is 7^1 .

For the prime factor 11, the highest power that can be included is 11^1 .

For our given numbers, the number of positive integers that divide $(315) \times (96) \times (550)$ and are divisible by 18 is:

$$(5 + 1) \times (1 + 1) \times (3 + 1) \times (1 + 1) \times (1 + 1) = 6 \times 2 \times 4 \times 2 \times 2 = 192.$$

Therefore, the number of positive integers that divide $(315) \cdot (96) \cdot (550)$ and are not divisible by 18 is $448 - 192 = 256$.

55. **Answer:** (4)

Let's assume the scores of Rahul, Priya and Sameer after rechecking be $3x$, $4x$ and $5x$, respectively.

Before rechecking, their scores were $3x - 15$, $4x - 15$ and $5x - 15$, respectively.

According to the given condition,

Rahul's score is one-fourth of the sum of Priya's and Sameer's scores:

$$3x - 15 = (4x - 15 + 5x - 15)/4$$

$$3x - 15 = (9x - 30)/4$$

$$12x - 60 = 9x - 30$$

$$12x - 9x = -30 + 60$$

$$3x = 30$$

$$x = 10$$

The difference between Sameer's score and Priya's score is: $5x - 4x = x = 10$

56. **Answer:** (2)

Let us suppose Lily can complete x units of work in a day and Hema can complete y units of work in a day.

Let the total work be 150 units.

Lily works alone until one-third of the job is completed. Lily does $\frac{1}{3}$ of 150 = 50 units.

Both Lily and Hema work together for 8 days. Hema works alone to finish the $\frac{2}{5}$ of the job.

Work that Lily and Hema did working together for 8 days = $150 - (50 + \frac{2}{5} \text{ of } 150) = 40$ units

$$8x + 8y = 40$$

$$x = 1.75y$$

$$14y + 8y = 40$$

$$22y = 40$$

$$y = \frac{20}{11} \text{ units per day}$$

$$\text{Required number of days} = \frac{150 \times 11}{20} = 82.5$$



57. **Answer:** (2)

In the first 6 performances, the average was 80. So, the score obtained in the first 6 performances = 480

After 7th performance, it was 83; so total score = $83 \times 7 = 581$

After 8th performance, it was 85; so total score = $85 \times 8 = 680$

Since she scores 130 points in Singing,

Her total score after Singing = $680 + 130 = 810$

Total number of performances = 9

Final average score = $810/9 = 90$

58. **Answer:** (3)

Let's assume the prize money received by Anand, Bipin, and Kiran in 2018 to be $8x$, $4x$, and $5x$, respectively.

Also assume the prize money received by them in 2023 to be $4y$, $5y$, and $7y$, respectively.

Given that Anand's prize money increased by 30% during the period 2018-2023,

New value for Anand's prize money = $8x + (30\% \text{ of } 8x) = 10.4x$

ATQ

$10.4x = 4y$

$\Rightarrow y = 2.6x$

Kiran's prize money in 2023 = $7y = 7(2.6x) = 18.2x$

Increase in Kiran's prize money = $18.2 - 5x = 13.2x$

Percentage increase in Kiran's prize money = $\frac{13.2}{5} \times 100 = 264\%$

59. **Answer:** (3)

As $f(x)$ is divisible by both $(x - 2)$ and $(x + 3)$,

$f(x) = (x - 2)(x + 3)q(x) + r(x)$, where $q(x)$ represents the quotient when $f(x)$ is divided by $(x - 2)(x + 3)$, and $r(x)$ is the remainder.

We're given that when $f(x)$ is divided by $(x - 2)$, the remainder is 6. So, $f(2) = 6$.

Similarly, when $f(x)$ is divided by $(x + 3)$, the remainder is -4, i.e. $f(-3) = -4$.

At $x = 2$, we get $f(2) = (2 - 2)(2 + 3)q(2) + r(2)$

$= 0 \times q(2) + r(2)$

$\Rightarrow r(2) = 6$

Similarly,

At $x = -3$, $f(-3) = (-3 - 2)(-3 + 3)q(-3) + r(-3)$

$= 0 \times q(-3) + r(-3)$

$\Rightarrow r(-3) = -4$

Now, the degree of the remainder equation will be less than that of the quotient equation.

Hence, the remainder equation will be linear of form $ax + b$.

i.e. $r(x) = ax + b$

$2a + b = 6 \dots(1)$

$-3a + b = -4 \dots(2)$

Solving (1) and (2), we get $a = 2$ and $b = 2$

Therefore, the remainder equation will be $ax + b = 2x + 2$.



60. **Answer:** (3)

Let the points accumulated from field goals, touchdowns, and successful extra-point conversions made by two players, Ajay and Vinay, be 'F', 'E', and P, respectively.

$$F + E + P = 53 \text{(1)}$$

$$P = 12 E \text{(2)}$$

$$E = F + 3 \text{(3)}$$

By using (1), (2) and (3), we get

$$E - 3 + E + 12E = 53$$

$$14E = 56$$

$$\text{So, } E = 4, F = 1 \text{ and } P = 48$$

Further, the ratio of the points scored by Ajay to that by Vinay was 3 : 5.

$$\text{Then, points scored by Ajay} = \frac{3}{8} \text{ of } 48 = 18$$

61. **Answer:** (1)

$$F = 6P^2 - 3Q^2 + R + 0.5S$$

The relationships are:

$$(i) 2P + Q \leq 5$$

$$(ii) 4P + 2Q + R \leq 10$$

$$(iii) 2P + 2Q + 3S \leq 29$$

For maximum F, Q must be minimum, i.e. Q = 0 (non-negative integer).

From equation (i), $2P + 0 \leq 5$

$$P = 2 < \frac{5}{2}$$

From equation (iii), $2 \times 2 + 4 \times 0 + 3S \leq 29$

$$3S \leq 25$$

$$S \leq 8.334$$

So, value of S = 8

So, F is maximum when S is maximum.

Required value of S = 8

Also, from (ii), $4P + 2Q + R \leq 10$

$$8 + 0 + R \leq 10$$

So, R = 2

The maximum value of $F = 6P^2 - 3Q^2 + R + 0.5S$

$$= 6(4) - 3(0) + 2 + 0.5(8)$$

$$= 24 + 6$$

$$= 30$$

62. **Answer:** (2)

From statement (I), the combined amount of the estate that the two younger children received can be determined, but not the individual amount received by either of them.

Thus, statement (I) alone is not sufficient.

Using statement (II),



Let x , y and z be the amounts received by the youngest child, the eldest child and the spouse, respectively.

So, according to the question:

$$x = 12,500 + y$$

$$x = z - 62,500$$

$$2x + y + z = 2,00,000$$

Three equations and three variables

So, statement (II) alone is sufficient.

63. **Answer:** (3)

Total cost of type X and type Y shirts = Rs. 4,750

Let the cost per shirt of type Y be ' x ' (in rupees).

Then, the cost per shirt of type X is $1.25x$ (25% more than Y).

Let the number of type Y shirts be ' y '.

Then, the number of type X shirts is $3y$.

The total cost can be expressed as:

$$(3y)(1.25x) + (y)(x) = 4,750$$

$$3.75xy + xy = 4,750$$

$$4.75xy = 4,750$$

$$xy = 4,750/4.75 = 1,000$$

$$\text{Cost of type X shirts} = (3y)(1.25x) = 3,750$$

$$\text{Cost of type Y shirts} = (y)(x) = 1,000$$

$$\text{Selling price of type X} = (1.20)(\text{Cost of X}) = 1.20(3,750) = 4,500$$

$$\text{Selling price of type Y} = (1.15)(\text{Cost of Y}) = 1.15(1,000) = 1,150$$

$$\text{Total selling price} = \text{Selling price of type X} + \text{Selling price of type Y} = 5,650$$

$$\text{Overall profit} = \text{Total selling price} - \text{Total cost} = 5,650 - 4,750 = 900$$

$$\text{Overall profit} = 900$$

64. **Answer:** (1)

Let the amount of money earned by Isha, Fatima and Garima in a week be e , f and g .

According to question,

Isha and Fatima together earn money in one week equal to the amount of money earned by Garima in four weeks.

$$\text{Then } e + f = 4g \text{(1)}$$

Further, the amount earned by Garima and Isha together is equal to the amount earned by Fatima in two weeks.

$$\text{Then } g + e = 2f \text{(2)}$$

By solving (1) and (2), we get

$$3f = 5g$$

$$f = 5g/3$$

Put the value in (1).

$$e = 4g - 5g/3 = 7g/3$$



So, the highest amount earner among the three is Isha, i.e. $7g/3$, and the lowest amount earner among the three is Garima, i.e. g .

Required ratio = $7 : 3$

65. **Answer:** (2)

Given: Speed of Amit = 12 km/hr

We know, Distance = Speed $\times$ Time

Distance travelled by Amit in 30 minutes = $12 \text{ kilometres per hour} \times 0.5 \text{ hours} = 6 \text{ kilometres}$

Further,

When Bimal starts cycling, Amit has already covered 6 kilometres.

Relative speed = $15 \text{ km/hr} - 12 \text{ km/hr} = 3 \text{ km/hr}$

Time taken by Bimal = Distance travelled by Amit in 30 minutes / Relative speed

$$\frac{6}{3} = 2 \text{ hr}$$

Therefore, Bimal will catch up with Amit after 2 hours of cycling.

Distance = Bimal's Speed $\times$ Time taken by Bimal

= $15 \text{ kilometres per hour} \times 2 \text{ hours}$

= 30 kilometres

So, Bimal will meet Amit at a distance of 30 kilometres from the starting point.

66. **Answer:** (1)

A beats B by 2 seconds and B beats C by 4 seconds.

Hence, A beats C by = $(2 + 4) \text{ seconds} = 6 \text{ seconds}$.

A beats C by a distance of 24 metres.

So, the time taken by C to cover 24 metres is 6 seconds.

$$\text{Speed of C} = \frac{24}{6} = 4 \text{ m/s}$$

Total distance = 100 metres

$$\text{Time taken by C} = \frac{100}{4} = 25 \text{ seconds}$$

We know that A beats C by 6 seconds, so time taken by A = $25 - 6 = 19 \text{ seconds}$.

67. **Answer:** (3)

Let's assume the original revenue generated by the company before the discount and surge in sales is 100 units.

After the 15% promotional discount, the price of the flagship product will be 85% of its original price ($100\% - 15\% = 85\%$).

With the 30% surge in unit sales, the number of units sold will increase by 30%.

The total revenue would be $(85\% \times 130\%) = 110.5\%$ of the original revenue.

$$\text{Percentage increase} = \frac{110.5 - 100}{100} \times 100 = 10.5$$



68. **Answer:** (2)

The total number of gift baskets that need to be prepared is 2,800. Initially, the plan is for each volunteer to make an equal number of gift baskets.

Let's assume the number of gift baskets each volunteer was supposed to make is 'x' and there were total of 'm' volunteers initially.

So, total number of gift baskets made by all volunteers would be 'mx'.

However, on the day of preparation, 15 volunteers are absent.

So, the number of volunteers who actually make the gift baskets is 'm - 15'.

According to the revised plan, each volunteer makes 24 gift baskets more than originally decided.

So, the new number of gift baskets made by each volunteer is 'x + 24'.

$$mx = 2,800 \text{ (Given)} \dots\dots\dots(1)$$

$$(m - 15)(x + 24) = mx$$

$$mx - 15x + 24m - 360 = mx$$

$$24m - 15x = 360$$

$$\text{From (1), } x = 2800/m$$

$$24m - 15(2800/m) = 360$$

$$24m^2 - 42,000 - 360m = 0 \dots\dots\dots(2)$$

The roots are $m = 50$ and $m = -35$ (rejected as volunteers cannot be in negative).

Hence, $m = 50$, $x = 56$

The new number of gift baskets made by each volunteer is ' $x + 24 = 80$ '.

69. **Answer:** (2)

Initial cost of the product: 'C' rupees

First-day price increase: 'x' %

First-day price decrease (due to promotion): 'y' %

Second-day price increase: 'x' %

Second-day price decrease (due to promotion): 'y' %

$$\text{After the first-day price increase, the cost becomes: } C + \frac{x}{100}C = \left(1 + \frac{x}{100}\right)C \text{ rupees} \dots\dots\dots(1)$$

$$\text{Then the price decrease by } y\%. \text{ Then the cost becomes: } \left(1 + \frac{x}{100}\right)C - \frac{y}{100}\left(1 + \frac{x}{100}\right)C$$

$$= \left(1 + \frac{x}{100}\right)C \left[1 - \frac{y}{100}\right] \text{ rupees}$$

After the second-day price increase, the cost becomes:

$$\left(1 + \frac{x}{100}\right) \left[\left(1 + \frac{x}{100}\right)C \left[1 - \frac{y}{100}\right]\right] = \left(1 + \frac{x}{100}\right)^2 \left[1 - \frac{y}{100}\right]C \text{ rupees}$$

Then the price decreases by y% again. So, the cost becomes:

$$\left(1 - \frac{y}{100}\right) \left(1 + \frac{x}{100}\right)^2 \left[1 - \frac{y}{100}\right]C = \left(1 + \frac{x}{100}\right)^2 \left[1 - \frac{y}{100}\right]^2 C \text{ rupees}$$

We are given that 'y' is 25.

$$\text{So, } \left(1 + \frac{x}{100}\right)^2 \left[1 - \frac{y}{100}\right]^2 C = \left(1 + \frac{x}{100}\right)^2 \left[1 - \frac{25}{100}\right]^2 C = \frac{9}{16} \left(1 + \frac{x}{100}\right)^2 C$$



Now, the product's cost returns to 'C' rupees.

$$\Rightarrow \frac{9}{16} \left(1 + \frac{x}{100}\right)^2 C = C$$

By solving, we get $x = 33.33$

Put the value of x in equation (1).

After increase, the price of the product becomes: $\left(1 + \frac{33.33}{100}\right)C = 133.33C$

70. **Answer:** (2)

Car price = Rs. 3,00,000

Down payment = Rs. 75,000

Remaining balance = Rs. 2,25,000

Number of instalments = 4

Rate = 8%

At the end of 4 months, amounts to be paid = $2,25,000 + \frac{2,25,000 \times 8 \times 4}{12 \times 100} = 2,25,000 + 6,000 = 2,31,000$ (1)

Further,

Let instalment amount = x

Total instalment amount to be paid = $4x + \frac{x \times 8 \times 3}{12 \times 100} + \frac{x \times 8 \times 2}{12 \times 100} + \frac{x \times 8 \times 1}{12 \times 100}$

$$= 4x + \frac{8x(3+2+1)}{12 \times 100} = 4x + \frac{48x}{12 \times 100} = 4.04x$$
(2)

By comparing (1) and (2), we get

$$4.04x = 2,31,000$$

$$\text{So, } x = 57,178.2 = \text{Rs. } 57,180 \text{ (approx.)}$$

71. **Answer:** (2)

Let's denote the number of students who presented projects in Biology as B, in Chemistry as C, and in Physics as P.

Based on the information provided:

B = 60 (students who presented projects in Biology)

C = 80 (students who presented projects in Chemistry)

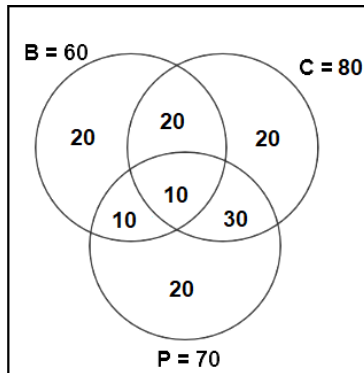
P = 70 (students who presented projects in Physics)

We are also given the following constraints:

At most 30 students presented projects in both Biology and Chemistry.

At most 40 students presented projects in both Chemistry and Physics.

At most 20 students presented projects in both Biology and Physics.



By using formula, we get

$$n(B \cup C \cup P) = n(B) + n(C) + n(P) - n(B \cap C) - n(C \cap P) - n(P \cap B) + n(B \cap C \cap P)$$

For the maximum value of students who presented projects in all three subjects will be:

$$130 = 60 + 80 + 70 - 30 - 40 - 20 + x$$

$$130 + 90 - 210 = x$$

$$\text{So, } x = 10$$

Hence, the maximum number of students who presented projects in all three subjects is 10.

72. **Answer:** (3)

$$\text{Here, } f(x - y) = \frac{f(x)}{f(y)}$$

The given condition is $f(a - 1) + f(a - 2) + f(a - 3) + \dots + f(a - n) = \frac{(3^n - 1)}{18}$.

$$\frac{f(a)}{f(1)} + \frac{f(a)}{f(2)} + \frac{f(a)}{f(3)} + \dots + \frac{f(a)}{f(n)} = \frac{3^n - 1}{18}$$

$$f(a) \left(\frac{1}{f(1)} + \frac{1}{f(2)} + \frac{1}{f(3)} + \dots + \frac{1}{f(n)} \right) = \frac{3^n - 1}{18} \quad \dots(1)$$

For $n = 1$,

$$f(a) \left(\frac{1}{f(1)} \right) = \frac{3^1 - 1}{18} = \frac{1}{9}$$

$$f(a) = \frac{1}{3}$$

Putting the value of $f(a)$ in (1), we get

$$\left(\frac{1}{f(1)} + \frac{1}{f(2)} + \frac{1}{f(3)} + \dots + \frac{1}{f(n)} \right) = \frac{3^n - 1}{18} \times 3 = \frac{3^n - 1}{6}$$

For $n = 2$, we get

$$\frac{1}{f(1)} + \frac{1}{f(2)} = \frac{3^2 - 1}{6} = \frac{8}{6}$$

$$\frac{1}{f(2)} = \frac{4}{3} - \frac{1}{3} = 1$$

$$\text{So, } f(2) = 1$$



For $n = 3$, we get

$$\left(\frac{1}{f(1)} + \frac{1}{f(2)} + \frac{1}{f(3)}\right) = \frac{3^3 - 1}{6} = \frac{13}{3}$$

$$\frac{1}{f(3)} = \frac{13}{3} - \frac{4}{3} = 3$$

$$f(3) = \frac{1}{3}$$

$$\text{Similarly, we get } f(4) = \frac{1}{9}$$

$$f(5) = \frac{1}{27} = \frac{1}{9} f(4)$$

73. **Answer:** (3)

Since there are 21 apples, out of which exactly 12 are bruised, number of not bruised apples = 9

For exactly 8 apples in the basket, which are not bruised:

So, after 5 days exactly 8 apples means 4 are bruised and 1 is not bruised.

The apples that are not bruised can be taken out on any of the 5 days.

Case (1):

The apples that are not bruised are taken out on the first day.

$$\text{Probability} = \frac{{}^9C_1 \times {}^{12}C_4}{{}^{21}C_1 \times {}^{20}C_4} = \frac{9}{21} \times \frac{12}{20} \times \frac{11}{19} \times \frac{10}{18} \times \frac{9}{17}$$

Case (2):

The apples that are not bruised are taken out on the second day.

$$\text{Probability} = \frac{{}^{12}C_1 \times {}^9C_1 \times {}^{11}C_3}{{}^{21}C_1 \times {}^{20}C_1 \times {}^{19}C_3} = \frac{12}{21} \times \frac{9}{20} \times \frac{11}{19} \times \frac{10}{18} \times \frac{9}{17}$$

Case (3):

The apples that are not bruised are taken out on the third day.

$$\text{Probability} = \frac{{}^{12}C_2 \times {}^9C_1 \times {}^{10}C_2}{{}^{21}C_2 \times {}^{19}C_1 \times {}^{18}C_2} = \frac{12}{21} \times \frac{11}{20} \times \frac{9}{19} \times \frac{10}{18} \times \frac{9}{17}$$

Case (4):

The apples that are not bruised are taken out on the fourth day.

$$\text{Probability} = \frac{{}^{12}C_3 \times {}^9C_1 \times {}^9C_1}{{}^{21}C_3 \times {}^{18}C_1 \times {}^{17}C_1} = \frac{12}{21} \times \frac{11}{20} \times \frac{10}{19} \times \frac{9}{18} \times \frac{9}{17}$$

Case (5):

The apples that are not bruised are taken out on the fifth day.

$$\text{Probability} = \frac{{}^{12}C_4 \times {}^9C_1}{{}^{21}C_4 \times {}^{17}C_1} = \frac{12}{21} \times \frac{11}{20} \times \frac{10}{19} \times \frac{9}{18} \times \frac{9}{17}$$



$$\text{Therefore, required probability} = 5 \times \left(\frac{9}{21} \times \frac{12}{20} \times \frac{11}{19} \times \frac{10}{18} \times \frac{9}{17} \right) = \frac{495}{2,261}$$

74. **Answer:** (2)

Given that:

$$10^{\text{th}} \text{ term of the arithmetic progression} = a + 9d$$

$$15^{\text{th}} \text{ term of the arithmetic progression} = a + 14d$$

$$25^{\text{th}} \text{ term of the arithmetic progression} = a + 24d$$

We are told that these terms form three consecutive terms of a geometric progression.

$$(a + 14d)^2 = (a + 9d)(a + 24d)$$

By solving, we get

$$a = -4d \text{ (d = 0 is rejected)}$$

Let's denote the common ratio of the geometric progression as 'r'.

$$\text{Then } r = \frac{a_{15}}{a_{10}} = \frac{10d}{5d} = 2$$

Now, we need to find the next term of the geometric progression, which is the n^{th} term of the arithmetic progression.

The next term of G.P. is expressed as:

$$(a + 24d) \times r = (20d) \times 2 = 40d \text{(1)}$$

Since the next term is the n^{th} term of the arithmetic progression, we have:

$$a_n = a + (n - 1)d \text{(2)}$$

By comparing (1) and (2), we get

$$-4d + (n - 1)d = 40d$$

$$\Rightarrow (n - 1)d = 44d$$

Since $d \neq 0$

So, $n = 45$ is the required value.

75. **Answer:** (1)

Here,

$$\int \frac{\beta x}{e^{2 \log x} + 1} dx$$

$$\Rightarrow \int \frac{\beta x}{x^2 + 1} dx$$

$$\Rightarrow \frac{\beta}{2} \log(x^2 + 1) + c_1$$

Now,

$$\int e^x \left\{ \alpha \log(e^{2 \log x} + 1) + \frac{\beta x}{e^{2 \log x} + 1} \right\} dx = \frac{\beta}{2} e^x \log(x^2 + 1)$$

$$\int e^x \left\{ \alpha \log(x^2 + 1) + \frac{\beta x}{x^2 + 1} \right\} dx = \frac{\beta}{2} e^x \log(x^2 + 1)$$

Consider the integral property given below.



$$\int e^x [f(x) + f'(x)] dx = e^x f(x)$$

Let

$$f(x) = \alpha \log(x^2 + 1)$$

$$f'(x) = \frac{\alpha \cdot 2x}{x^2 + 1}$$

To use the above integral property,

$$\alpha \frac{2x}{x^2 + 1} = \frac{\beta x}{x^2 + 1}$$

$$2\alpha = \beta$$

$$\alpha = \frac{\beta}{2}$$

Let us check the integral value with R.H.S.

$$\int e^x \left[\alpha \log(x^2 + 1) + \frac{2\alpha x}{x^2 + 1} \right] dx = e^x \alpha \log(x^2 + 1) + c$$

Compare this result with R.H.S. of the given equation.

$$e^x \alpha \log(x^2 + 1) + c = \frac{\beta}{2} e^x \log(x^2 + 1) + c$$

$$\alpha = \frac{\beta}{2}$$

It is possible when,

$$\beta = 2, \alpha = 1$$

Hence, this is the required solution.

76. **Answer:** (4)

$$\text{We know that } \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}.$$

$$\text{So, } m = \sin^2\left(\frac{\pi}{6}\right) + \sin^4\left(\frac{\pi}{6}\right) + \sin^6\left(\frac{\pi}{6}\right) + \dots + \infty$$

$$m = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^{2k} = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$$

$$\text{Now, function } f(x) = \frac{x}{2^3} + 2^{\frac{1}{3}(1-x)}$$

As we know, A.M. $\geq$ G.M.

$$\frac{\frac{x}{2^3} + 2^{\frac{1-x}{3}}}{2} \geq \left(2^{\frac{x}{3} + \frac{1-x}{3}}\right)^{\frac{1}{2}}$$

$$\Rightarrow f(x) \geq 2^{\frac{7}{6}}$$



$$\frac{x}{2^3} + \frac{2^{\frac{1}{3}}}{x}$$

Now, $f(x) =$

$$= \frac{x}{2^3} + 2^{\frac{1}{3}} \cdot 2^{-\frac{x}{3}}$$

$$\therefore f'(x) = \frac{x}{2^3} \cdot \log 2 \cdot \frac{1}{3} + 2^{\frac{1}{3}} \cdot 2^{-\frac{x}{3}} \cdot \log 2 \cdot \left(-\frac{1}{3}\right)$$

$$= \frac{1}{3} \log 2 \left[\frac{\frac{x}{2^3} - \frac{2^{\frac{1}{3}}}{2^{\frac{x}{3}}}}{2^{\frac{x}{3}}} \right]$$

$$= \frac{1}{3} \log 2 \left[\frac{\frac{2x}{2^3} - \frac{1}{2^{\frac{x}{3}}}}{\frac{x}{2^3}} \right]$$

Therefore, $f'(x) = 0$ at $x = \frac{1}{2}$

Since at $x = \frac{1}{2}$, $f(x) = 2^{\frac{7}{6}}$ which is minimum.

Now, $f'(x) > 0$ when

$$\frac{1}{3} \log 2 \left[\frac{\frac{2x}{2^3} - \frac{1}{2^{\frac{x}{3}}}}{\frac{x}{2^3}} \right] > 0$$

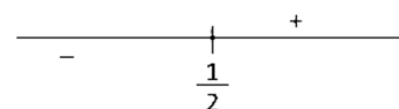
$$\Rightarrow \frac{2x}{2^3} > \frac{1}{2^{\frac{x}{3}}}$$

$$\Rightarrow \frac{2x}{3} > \frac{1}{3}$$

$$\Rightarrow x > \frac{1}{2}$$

Similarly, $f'(x) < 0$ when $x < \frac{1}{2}$

If we put it in the number line, we get this:

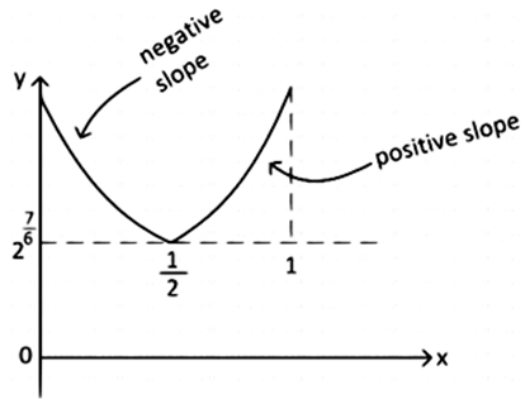


We know, $f'(x)$ represents the slope of curve $f(x)$ and it is negative when $x < \frac{1}{2}$, positive when

$x > \frac{1}{2}$ and zero when $x = \frac{1}{2}$.



∴ Graph of $f(x)$ is:



From the graph, we can see the value of $f(x)$ is maximum either at $x = 0$ or $x = 1$ in the range $x \in [0, 1]$.

$$\therefore f(0) = 2^0 + 2^{-\frac{1}{3}} = 1 + 2^{-\frac{1}{3}}$$

$$f(1) = 2^{-\frac{1}{3}} + 2^0 = 1 + 2^{-\frac{1}{3}}$$

∴ We get maximum value at $x = 0$ and $x = 1$.

So, options 1, 2, 3 and 5 are true and option 4 is not true.

77. **Answer:** (1)

$$af(x) + bf\left(\frac{1}{x}\right) = x^2 + \frac{1}{x^2}$$

$$af\left(\frac{1}{x}\right) + bf(x) = x^2 + \frac{1}{x^2}$$

Subtracting

$$(a - b)f(x) + (b - a)f\left(\frac{1}{x}\right) = 0$$

$$(a - b)f(x) = (a - b)f\left(\frac{1}{x}\right)$$

$$f(x) = f\left(\frac{1}{x}\right)$$

$$(a + b)f(x) = x^2 + \frac{1}{x^2}$$

$$f(x) = \frac{1}{a + b} \left[x^2 + \frac{1}{x^2} \right]$$

$$f'(x) = \frac{1}{a + b} \left[2x - \frac{2}{x^3} \right]$$

$$f'(2) = \frac{1}{a + b} \left[4 - \frac{1}{4} \right] = \frac{15}{4(a + b)}$$



78. **Answer:** (4)

By using identity:

$$AA^{-1} = I$$

$A = AI = A$; where A is any matrix and I is an identity matrix.

The given expression is $2P - Q - P(P - Q)^{-1}P + Q(P - Q)^{-1}Q$.

$$\begin{aligned} &= 2P - Q - (P - Q)^{-1}(P \cdot P - Q \cdot Q) \\ &= 2P - Q - (P - Q)^{-1}(P^2 - Q^2) \\ &= 2P - Q - (P - Q)^{-1}(P - Q)(P + Q) \\ &= 2P - Q - I(P + Q) \\ &= 2P - Q - (P + Q) \\ &= 2P - Q - P - Q \\ &= P - 2Q \end{aligned}$$

79. **Answer:** (4)

By using the property of determinant, we get

$$\begin{aligned} |A| &= \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 - 1 & b^3 - 1 & c^3 - 1 \end{vmatrix} \\ &= \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} - \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ 1 & 1 & 1 \end{vmatrix} \\ &= abc \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} - \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ 1 & 1 & 1 \end{vmatrix} \\ &= abc \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} - \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} (abc - 1) \\ &= \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = 0 \end{aligned}$$

So, $abc = 1$ is the required value.



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80. **Answer:** (4)

Given: $M = \{1, 2, 3, 4\}$ and $N = \{p, q\}$

Relation from M to N is given as: $M \times N = \{(1, p), (1, q), (2, p), (2, q), (3, p), (3, q), (4, p), (4, q)\}$

Number of elements in relation M to N = 8

$\therefore$ Total number of relations obtained = 2^8

Also, the total number of functions from M to N is given as r^n , where r is the number of elements in N and n is the number of elements in M.

Total number of functions = 2^4

Probability of having a function from the relation = $\frac{2^4}{2^8} = \frac{1}{16}$

Now, required probability of not having a function = $1 - \frac{1}{16} = \frac{15}{16}$

----- **END OF PAPER** -----



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